THE AESTHETIC GENOME

Fashion Retail Sales Analytics

PROJECT DOCUMENTATION

Cover-page item	Details
Project Title	The Aesthetic Genome — Fashion Retail Sales Analytics
Name	Sivani Paladugu
Batch / Class	BN010
Project Type	Capstone Project
Tools Used	SQL; Python (pandas, NumPy, matplotlib, seaborn); Power BI; DAX,
Submission Date	12th October 2026

2. Project Overview

The Aesthetic Genome is an end-to-end data analytics project for a 2024 fashion-retail transaction dataset. It moves from raw-data quality assessment and SQL cleaning through Python analysis, Power BI data modelling, DAX measures and a five-page management dashboard.

The analytical story is organized around three questions: What happened to revenue and order behaviour? What observable product, pricing, discount and customer patterns are associated with the result? What should management test or improve next?

Project component	Documented scope
Data	50,800-row raw file; 50,000-row cleaned file; 49,850-row Python valid-order population.
SQL	MySQL database setup, data cleaning, validation, KPI analysis, category/city/customer/discount analysis, monthly/MoM trends and window functions.
Python	Inspection, descriptive statistics, correlations, outlier analysis and the final nine-chart business story.
Power BI	Five report pages: Executive Overview; Sales & Growth; Product, Pricing & Basket Economics; Customer & Action Center; Detailed Analysis & Drilldown.
Business use	Support cautious decisions around discount depth, basket-building, customer segmentation and source-data quality.

3. Introduction

3.1 Background

The supplied files describe fashion-retail orders with product, aesthetic style, colour, lifestyle, price, discount, sales, profit, city, payment and customer fields. The notebook identifies the dataset as The Aesthetic Genome and uses 2024 transaction data.

3.2 Problem Statement

The raw data contains duplicate Order_ID records, missing descriptive values, negative quantities and zero-value records. Without a documented preparation and analytical workflow, revenue, order value, discount and customer interpretations could be inconsistent.

3.3 Need for the Project

- Create a reproducible data-cleaning process before analysis.
- Measure revenue, order value, basket size, category, city, customer and discount patterns together.
- Distinguish a typical order from legitimate high-value baskets.
- Make findings available through SQL, Python and Power BI for both technical review and business communication.

3.4 Objectives

1. Inspect and clean the raw transaction data using SQL.
2. Validate the cleaned dataset and define the valid-order population for Python analysis.
3. Analyse KPIs, product/category, geography, customer, discount and time patterns.
4. Use nine business visualizations to explain what happened, what is associated with the result, and what should be tested.
5. Build a Power BI model and five-page dashboard while preserving the project's existing page structure.
6. Translate supported findings into cautious recommendations and explicit limitations.

3.5 Scope

In scope are SQL cleaning and analysis, Python inspection/statistics/visualization, Power BI modelling, DAX-bound KPIs, dashboard documentation, business insights and recommendations. Machine learning, forecasting, deployment, APIs and cloud platforms are outside the supplied project scope and are not introduced.

4. Dataset Description

4.1 Data Source

File / artifact	Use in the project
new_raw_dataset.csv	Raw transaction input.
new_cleaned_data_power.csv	Cleaned output used by Power BI / analysis.
Data_Cleaning_using_SQL.sql	Database creation, table schema, cleaning and deduplication.
Data_Analysis_SQL.sql	KPI, category, geography, customer, discount, monthly and advanced analysis.
python_data_visualization_9_charts.ipynb	Python inspection, statistics, visualizations and final story.
major_project.pbix	Power BI data model, DAX-bound visuals, five dashboard pages and report theme.

4.2 Dataset Overview

Both CSV files are transaction-oriented flat files with 15 fields. The raw file contains 50,800 records and 50,000 distinct Order_ID values. The cleaned file contains 50,000 records with a unique Order_ID. The Python notebook creates a valid-order population by excluding Quantity = 0 rows.

4.3 Dataset Size

Metric	Raw	Cleaned	Python valid
Rows	50,800	50,000	49,850
Columns	15	15	15 + Order_Month + Month_Num
Distinct Order_ID	50,000	50,000	49,850
Distinct Customer_ID	8,969	8,969	8,969
Full duplicate rows	712	0	0
Null cells	4,839	0	0
Date coverage	2024-01-01 to 2024-12-31	2024-01-01 to 2024-12-31	2024-01-01 to 2024-12-31

4.4 Data Structure

The transaction grain is one order-level row in the supplied flat files. Identifier fields describe orders and customers; categorical fields describe products, styles, colours, café preferences, cities and payment methods; numeric fields describe quantity, price, discount, sales and profit.

4.5 Data Dictionary

Field	Type	Role	Observed values / quality note
Order_ID	Text	Order key	50,000 unique in cleaned file
Order_Date	Date text	Transaction timestamp	Parse before time analysis
Customer_ID	Text	Customer key	8,969 distinct
Product_Category	Categorical	Product grouping	6 categories

Field	Type	Role	Observed values / quality note
Fabric	Categorical	Material descriptor	13 cleaned values; Unknown observed
Aesthetic_Style	Categorical	Style grouping	4 styles
Color_Palette	Categorical	Colour grouping	8 values
Preferred_Cafe_Style	Categorical	Lifestyle preference	9 cleaned values; Unspecified observed
Quantity	Integer	Units/order	0–4 cleaned; 1–4 valid
Unit_Price	Decimal	Price measure	0–250 cleaned; 25–250 valid
Discount_Percent	Decimal fraction	Discount rate	0, .05, .10, .15, .20
Total_Sales	Decimal	Revenue measure	0–999.36 cleaned
Profit	Decimal	Profit field	Exactly 45% of sales in valid data
City	Categorical	Geography	9 cleaned values, including Unknown
Payment_Method	Categorical	Payment channel	4 methods

5. Data Preparation

5.1 Data Quality Issues

Issue	Raw evidence	Treatment / effect
Duplicate records	712 full-row duplicates; 800 duplicate Order_ID rows	Deduplicate by Order_ID in SQL.
Missing text	Fabric 2,300; café style 1,012; City 1,527 nulls	Impute and standardize labels.
Negative quantity	150 rows at -1	Set to 0 before valid-order filtering.
Zero quantity	150 rows in cleaned file	Excluded from Python valid population.
Zero Unit_Price	100 cleaned rows	Retained in cleaned file; requires source investigation.
Date stored as text	Profiler warning on both CSV files	Parse to datetime before time analysis.
Profit construction	Profit / Total_Sales = 0.45	Treat profit as mechanically derived, not an independent driver.

5.2 SQL Data Cleaning

The supplied cleaning script creates capstone_aesthetic_db, defines raw_aesthetic_sales, applies text and numeric updates, creates cleaned_aesthetic_sales with a window-function deduplication step, and verifies the cleaned record count.

```sql
CREATE TABLE raw_aesthetic_sales (
    Order_ID VARCHAR(50),
    Order_Date VARCHAR(50),
    Customer_ID VARCHAR(50),
    Product_Category VARCHAR(50),
    Fabric VARCHAR(50),
    Aesthetic_Style VARCHAR(50),
    Color_Palette VARCHAR(100),
    Preferred_Cafe_Style VARCHAR(100),
    Quantity INT,
    Unit_Price DECIMAL(10,2),
    Discount_Percent DECIMAL(5,2),
    Total_Sales DECIMAL(10,2),
    Profit DECIMAL(10,2),
    City VARCHAR(50),
    Payment_Method VARCHAR(50)
);
```

5.3 Cleaning Steps

7. Trim whitespace from text columns.
8. Standardize selected text casing for Fabric, Product_Category and City.
9. Replace missing/blank text values with documented placeholders.
10. Convert negative quantities to zero.
11. Recalculate Total_Sales and Profit.
12. Use ROW_NUMBER() over Order_ID to keep one row per order.
13. Verify the final record count.

5.4 Validation

Check	Result
Cleaned rows	50,000
Full duplicate rows	0
Order_ID uniqueness	Yes; 50,000 distinct
Null cells	0
Python valid rows	49,850
Valid customers	8,969
Valid revenue	15,428,005.95
Valid profit	6,942,603.24
Valid AOV	309.49
Valid units	124,891

5.5 Power Query Data Cleaning and Transformation

Power Query was used in Power BI to prepare the data before it was loaded into the data model. This stage complements the SQL data-cleaning process and supports the preparation of data for reporting and visualization.

The transformations performed in Power Query should be documented based on the actual steps applied in the Power Query Editor, including data type changes, column transformations, missing-value handling, or other preparation steps where applicable.

These transformations prepared the data for use in the Power BI data model and dashboard.

5.6 Final Dataset

The cleaned CSV is suitable for Power BI and further analysis, subject to the known label and population differences documented later. The Python report uses Quantity > 0 as its valid-order rule. The Power BI model exposes Is_Valid_Order for filtering.

6. SQL Analysis

The main report presents selected SQL analyses using the requested

QUESTION → QUERY → OUTPUT → OBSERVATION → BUSINESS MEANING structure.

Complete source scripts are included in the Appendix as project artifacts.

6.1 SQL Database Setup

QUESTION

What database and table structure supports the analysis?

QUERY

```sql
CREATE TABLE raw_aesthetic_sales (
    Order_ID VARCHAR(50),
    Order_Date VARCHAR(50),
    Customer_ID VARCHAR(50),
    Product_Category VARCHAR(50),
    Fabric VARCHAR(50),
    Aesthetic_Style VARCHAR(50),
    Color_Palette VARCHAR(100),
    Preferred_Cafe_Style VARCHAR(100),
    Quantity INT,
    Unit_Price DECIMAL(10,2),
    Discount_Percent DECIMAL(5,2),
    Total_Sales DECIMAL(10,2),
    Profit DECIMAL(10,2),
    City VARCHAR(50),
    Payment_Method VARCHAR(50)
);
```

OUTPUT

Output	Result
Database	capstone_aesthetic_db
Raw table	raw_aesthetic_sales
Cleaned table	cleaned_aesthetic_sales
Source columns	15

OBSERVATION

The script establishes a reproducible MySQL staging and cleaned-table workflow.

BUSINESS MEANING

A defined table structure separates raw ingestion from cleaned analysis and makes the transformation steps auditable.

6.2 Data Validation

QUESTION

How much data is present before and after cleaning?

QUERY

```
-- STEP 6: VERIFY CLEANED DATASET RECORD COUNT

SELECT COUNT(*) AS total_cleaned_records FROM cleaned_aesthetic_sales;

-- Re-enable Safe Updates
SET SQL_SAFE_UPDATES = 1;

SELECT *
FROM cleaned_aesthetic_sales
WHERE Order_Date IS NULL;
```

OUTPUT

Validation metric	Raw	Cleaned	Python valid
Rows	50,800	50,000	49,850
Order_ID distinct	50,000	50,000	49,850
Full duplicates	712	0	0
Null cells	4,839	0	0
Zero quantity	0	150	Excluded

OBSERVATION

The cleaned table has one row per Order_ID and no full-row duplicates or null cells; the Python population removes the 150 zero-quantity rows.

BUSINESS MEANING

The population must be labelled so SQL/Power BI totals are not compared directly with Python AOV or order-count metrics without reconciliation.

6.3 KPI Analysis

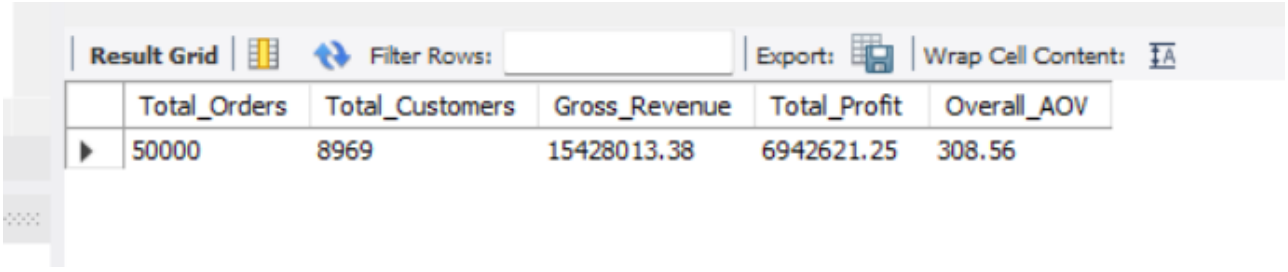
QUESTION

What are the overall orders, customers, revenue, profit and AOV?

QUERY

```
SELECT COUNT(DISTINCT Order_ID) AS Total_Orders,
       COUNT(DISTINCT Customer_ID) AS Total_Customers,
       ROUND(SUM(Total_Sales),2) AS Gross_Revenue,
       ROUND(SUM(Profit),2) AS Total_Profit,
       ROUND(AVG(Total_Sales),2) AS Overall_AOV
FROM cleaned_aesthetic_sales;
```

OUTPUT



Total_Orders	Total_Customers	Gross_Revenue	Total_Profit	Overall_AOV
50000	8969	15428013.38	6942621.25	308.56

OBSERVATION

Revenue and profit are unchanged by the 150 zero rows, but SQL AOV includes them and is therefore lower than the Python valid-order AOV of 309.49.

BUSINESS MEANING

AOV is population-sensitive; dashboard and report users should know whether invalid/zero rows are included.

6.4 Product and Category Analysis

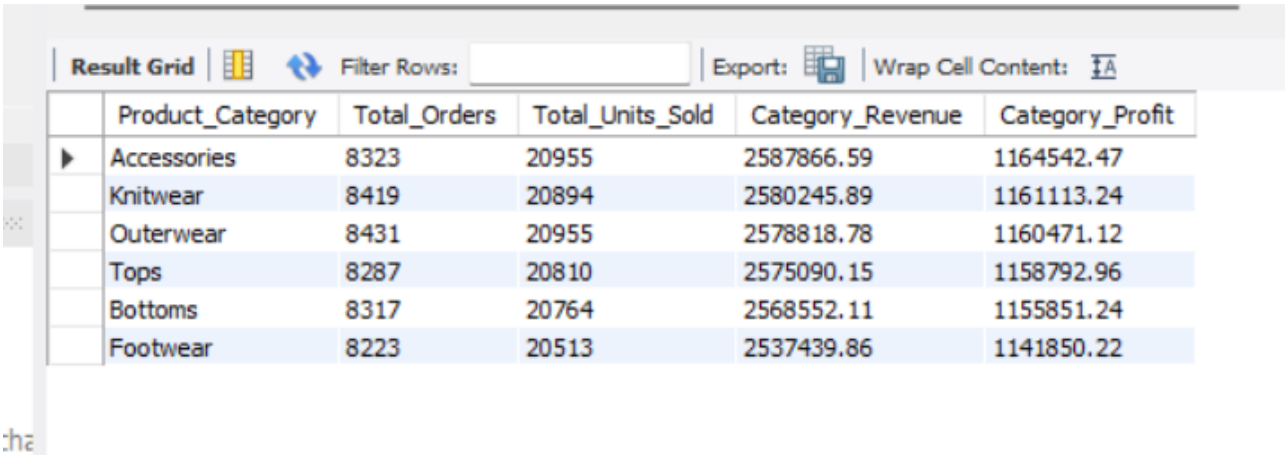
QUESTION

Which product categories contribute the most revenue?

QUERY

```
SELECT Product_Category, COUNT(Order_ID) AS Total_Orders,
       SUM(Quantity) AS Total_Units_Sold,
       ROUND(SUM(Total_Sales),2) AS Category_Revenue,
       ROUND(SUM(Profit),2) AS Category_Profit
FROM cleaned_aesthetic_sales
GROUP BY Product_Category
ORDER BY Category_Revenue DESC;
```

OUTPUT


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product_Category	Total_Orders	Total_Units_Sold	Category_Revenue	Category_Profit
Accessories	8323	20955	2587866.59	1164542.47
Knitwear	8419	20894	2580245.89	1161113.24
Outerwear	8431	20955	2578818.78	1160471.12
Tops	8287	20810	2575090.15	1158792.96
Bottoms	8317	20764	2568552.11	1155851.24
Footwear	8223	20513	2537439.86	1141850.22

OBSERVATION

Accessories is highest and Footwear is lowest, but the revenue range is narrow relative to the total.

BUSINESS MEANING

The evidence supports protecting a broad category base rather than treating one category as the only growth engine; profit rankings mirror sales because profit is derived at 45%.

6.5 Geographic Analysis

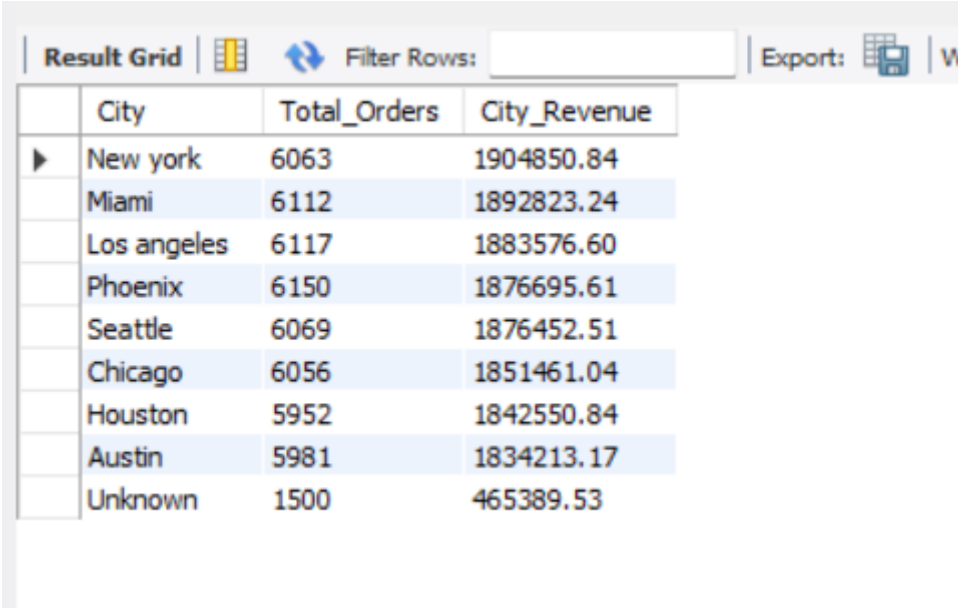
QUESTION

Which cities generate the most revenue?

QUERY

```
SELECT City, COUNT(Order_ID) AS Total_Orders,
       ROUND(SUM(Total_Sales),2) AS City_Revenue
FROM cleaned_aesthetic_sales
GROUP BY City
HAVING SUM(Total_Sales) > 5000.00
ORDER BY City_Revenue DESC;
```

OUTPUT



City	Total_Orders	City_Revenue
New york	6063	1904850.84
Miami	6112	1892823.24
Los angeles	6117	1883576.60
Phoenix	6150	1876695.61
Seattle	6069	1876452.51
Chicago	6056	1851461.04
Houston	5952	1842550.84
Austin	5981	1834213.17
Unknown	1500	465389.53

OBSERVATION

New York leads the named cities and Austin is lowest among the named cities; the spread is modest.

BUSINESS MEANING

Geographic targeting should be cautious because Unknown City records reduce the precision of city-level comparisons.

6.6 Customer Analysis

QUESTION

How does the supplied fixed-spend customer tiering classify customers?

QUERY

```
SELECT Customer_ID, COUNT(Order_ID) AS Lifetime_Orders,
       ROUND(SUM(Total_Sales),2) AS Lifetime_Spend,
       CASE WHEN SUM(Total_Sales) >= 500 THEN 'High-Value VIP'
            WHEN SUM(Total_Sales) BETWEEN 200 AND 499.99 THEN 'Mid-Tier Customer'
            ELSE 'Casual Shopper' END AS Customer_Tier
FROM cleaned_aesthetic_sales
GROUP BY Customer_ID;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Customer_ID	Lifetime_Orders	Lifetime_Spend	Customer_Tier
CUST-1000	6	1348.11	High-Value VIP
CUST-1001	6	1644.54	High-Value VIP
CUST-1002	7	1731.18	High-Value VIP
CUST-1003	3	868.87	High-Value VIP
CUST-1004	5	1899.14	High-Value VIP
CUST-1005	6	1242.40	High-Value VIP
CUST-1006	7	2628.41	High-Value VIP
CUST-1007	4	624.57	High-Value VIP
CUST-1008	7	2681.58	High-Value VIP
CUST-1009	7	2852.67	High-Value VIP
CUST-1010	5	1549.27	High-Value VIP
CUST-1011	6	1182.78	High-Value VIP
CUST-1012	2	791.04	High-Value VIP
CUST-1013	6	1661.97	High-Value VIP
CUST-1014	5	706.89	High-Value VIP
CUST-1015	6	1724.88	High-Value VIP
CUST-1016	7	2133.41	High-Value VIP
CUST-1017	6	1898.19	High-Value VIP
CUST-1018	6	2570.47	High-Value VIP
CUST-1019	8	2299.63	High-Value VIP
CUST-1020	3	1676.80	High-Value VIP
CUST-1021	4	787.29	High-Value VIP

OBSERVATION

The fixed $500 rule places 8,434 of 8,969 customers in High-Value VIP.

BUSINESS MEANING

This supports the project's note that a fixed $500 VIP threshold is too broad for prioritization; percentile-based tiers and repeat-purchase behaviour are more useful next steps.

6.7 Discount Analysis

QUESTION

Do discount bands show a visible basket-volume lift?

QUERY

```
SELECT CASE WHEN Discount_Percent = 0 THEN '0% (Full Price)'
            WHEN Discount_Percent BETWEEN 0.01 AND 0.10 THEN '1% - 10% Discount'
            ELSE 'Over 10% Discount' END AS Discount_Band,
       COUNT(Order_ID) AS Total_Orders,
       ROUND(AVG(Quantity),2) AS Avg_Units_Per_Order,
       ROUND(SUM(Total_Sales),2) AS Total_Revenue
FROM cleaned_aesthetic_sales
GROUP BY Discount_Band;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Discount_Band	Total_Orders	Avg_Units_Per_Order	Total_Revenue
0% (Full Price)	10011	2.48	3382274.93
1% - 10% Discount	19925	2.51	6377751.00
Over 10% Discount	20064	2.50	5667987.45

OBSERVATION

Average units per order is approximately 2.5 in all three bands. The notebook's finer five-level view shows AOV falling from 338.77 at 0% to 275.93 at 20%.

BUSINESS MEANING

Higher discount levels are associated with lower order value without a visible increase in units per order. This is an observational finding, not a causal claim.

6.8 Monthly / MoM Analysis

QUESTION

How does revenue change month over month?

QUERY

```
WITH MonthlySales AS (
 SELECT DATE_FORMAT(Order_Date,'%Y-%m') AS Sales_Month,
        ROUND(SUM(Total_Sales),2) AS Current_Month_Revenue
 FROM cleaned_aesthetic_sales
 GROUP BY DATE_FORMAT(Order_Date,'%Y-%m')
), MonthlyGrowth AS (
 SELECT Sales_Month, Current_Month_Revenue,
        LAG(Current_Month_Revenue) OVER (ORDER BY Sales_Month) AS Previous_Month_Revenue
 FROM MonthlySales
)
SELECT Sales_Month, Current_Month_Revenue, Previous_Month_Revenue,
       ROUND(((Current_Month_Revenue-Previous_Month_Revenue)/Previous_Month_Revenue)*100,2) AS MoM_Growth_Pct
FROM MonthlyGrowth;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Sales_Month	Current_Month_Revenue	Previous_Month_Revenue	MoM_Growth_Pct
NULL	767307.10	NULL	NULL
2024-01	1230633.23	767307.10	60.38
2024-02	1179391.02	1230633.23	-4.16
2024-03	1244397.59	1179391.02	5.51
2024-04	1232880.34	1244397.59	-0.93
2024-05	1249057.34	1232880.34	1.31
2024-06	1176946.78	1249057.34	-5.77
2024-07	1241811.49	1176946.78	5.51
2024-08	1256062.80	1241811.49	1.15
2024-09	1204798.28	1256062.80	-4.08
2024-10	1227418.63	1204798.28	1.88
2024-11	1174445.24	1227418.63	-4.32
2024-12	1242863.54	1174445.24	5.83

OBSERVATION

Revenue oscillates within a relatively narrow 2024 range; the largest displayed decline is June and the highest month is May.

BUSINESS MEANING

The evidence supports stable revenue with month-to-month variation, not a sustained growth or decline claim.

6.9 Advanced SQL Analysis

QUESTION

Which product categories rank within each aesthetic style?

QUERY

```
WITH RankedCategories AS (
 SELECT Aesthetic_Style, Product_Category,
        ROUND(SUM(Total_Sales),2) AS Category_Revenue,
        DENSE_RANK() OVER (PARTITION BY Aesthetic_Style
                           ORDER BY SUM(Total_Sales) DESC) AS Category_Rank
 FROM cleaned_aesthetic_sales
 GROUP BY Aesthetic_Style, Product_Category
)
SELECT * FROM RankedCategories WHERE Category_Rank <= 3;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Aesthetic_Style	Product_Category	Category_Revenue	Category_Rank
Coastal Minimalist	Bottoms	652966.73	2
Coastal Minimalist	Footwear	652139.92	3
Coastal Minimalist	Outerwear	663430.68	1
Streetwear Utility	Knitwear	646194.26	2
Streetwear Utility	Outerwear	642283.51	3
Streetwear Utility	Tops	658634.25	1
Urban Monochrome	Bottoms	641406.21	3
Urban Monochrome	Knitwear	644743.08	2
Urban Monochrome	Tops	654573.75	1
Vintage Academic	Accessories	662078.00	1
Vintage Academic	Outerwear	649321.47	3
Vintage Academic	Tops	649761.66	2

OBSERVATION

The query demonstrates within-style ranking; the leading category changes by style rather than one category dominating every style.

BUSINESS MEANING

Window functions support targeted drilldown without turning a modest ranking difference into an unsupported strategic conclusion.

6.10 SQL Findings

Finding	Evidence-based conclusion
Revenue base	Category and named-city revenue are broadly balanced.
Customer tiers	The fixed $500 VIP rule is broad; percentile-based tiers would support better prioritization.
Discounts	Higher discount bands do not show a clear units/order lift; finer levels show lower AOV at deeper discounts.
Time	Revenue is stable within a narrow band with alternating MoM increases and decreases.
Advanced analysis	CTEs, DENSE_RANK(), LAG() and running totals are used for useful business questions.

7. Python Analysis

Python is presented as Inspection → Statistics → Nine Business Visualizations → Findings. The report shows only short explanatory snippets and the final nine chart slots. The complete notebook remains a project artifact.

7.1 Python Data Loading and Inspection

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('new_cleaned_data_power.csv', parse_dates=['Order_Date'])
df = df[df['Quantity'] > 0].copy()
df['Order_Month'] = df['Order_Date'].dt.to_period('M').astype(str)
```

The supplied notebook currently references an Excel filename, while the attached cleaned artifact is CSV. The snippet above shows the equivalent file-format adaptation needed to run against the supplied CSV; no new analytical method is introduced.

7.2 Data Quality Validation

Check	Python valid result
Rows	49,850
Null cells	0
Full duplicates	0
Unknown City	1,499
Unknown Fabric	1,478
Mean Quantity	2.51
Median Total_Sales	247.46
Mean Total_Sales	309.49

7.3 Descriptive Statistics

Measure	Mean	Median	Minimum	Maximum
Quantity	2.51	3.00	1	4
Unit_Price	137.22	137.12	25.00	250.00
Discount_Percent	0.10	0.10	0.00	0.20
Total_Sales	309.49	247.46	20.01	999.36
Profit	139.27	111.35	9.00	449.71

7.4 Statistical Analysis

Analysis	Result	Interpretation
Price–quantity Pearson test	r ≈ 0.0037; p ≈ 0.40	No clear observed linear relationship between unit price and quantity in this dataset.
Unit_Price vs Total_Sales	r ≈ 0.69	Order value is positively associated with unit price; quantity forms visible bands.
Discount vs Total_Sales	r ≈ -0.11	Weak negative association; not causal evidence.

Analysis	Result	Interpretation
Profit vs Total_Sales	r = 1.00	Profit is mechanically derived at approximately 45% of sales.
Outlier threshold	887.5825	534 valid orders exceed the IQR upper fence; all are Quantity = 4.

7.5 Nine Business Visualizations

Act	Chart	Business question
Act 1	1. Monthly Revenue & Profit Trend	Is the business growing or remaining stable?
Act 2	2. Revenue by Category	Is revenue concentrated in one product category?
Act 3	3. Revenue by City	Is revenue geographically concentrated?
Act 4	4. Order Value Distribution	What does a typical order look like?
Act 5	5. High-Value Order / Outlier Analysis	Are high-value orders errors or legitimate baskets?
Act 6	6. Observed Price–Demand Relationship	Does unit price show an observed relationship with quantity?
Act 7	7. Discount vs Basket Volume / Order Value	Are higher discounts associated with higher basket volume?
Act 8	8. Correlation Heatmap	Which numerical variables move together?
Act 9	9. Customer Segment Contribution	Where should customer-retention effort be focused?

1. Monthly Revenue & Profit Trend

BUSINESS QUESTION

Is the business growing or remaining stable?

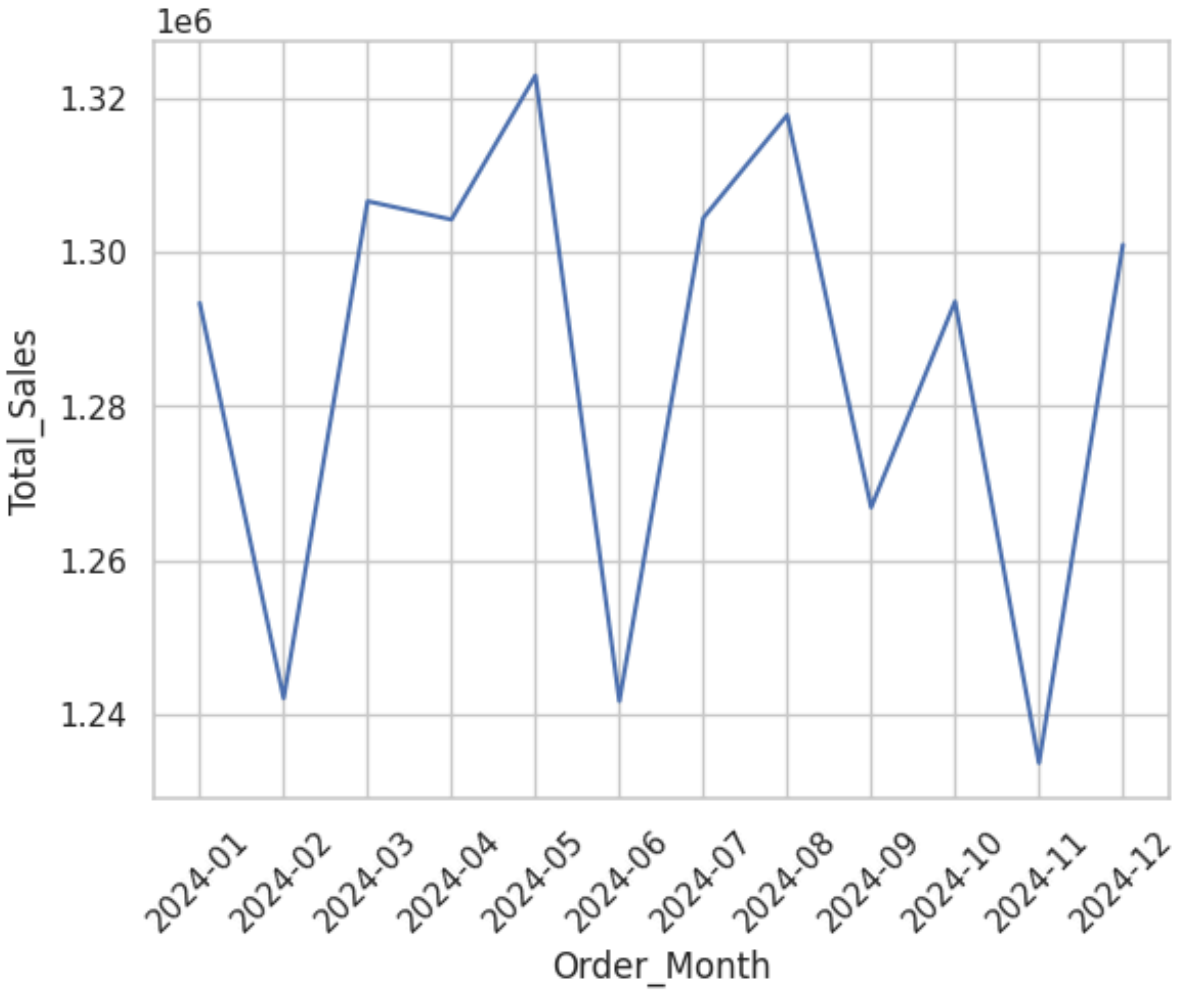


Figure 7.1: Monthly total-sales output from the supplied notebook

FINDING

Revenue remains within approximately 1.23M–1.32M across 2024, supporting a stable-revenue interpretation rather than a sustained-growth claim.

BUSINESS MEANING

The supplied Python image plots monthly Total_Sales. Profit is mechanically derived at approximately 45% of sales in the valid data, so it follows the same direction; no separate Python profit image is present in the supplied notebook.

DECISION / IMPLICATION

Use the result as a decision signal to test or investigate, not as proof of causality.

2. Revenue by Category

BUSINESS QUESTION

Is revenue concentrated in one product category?

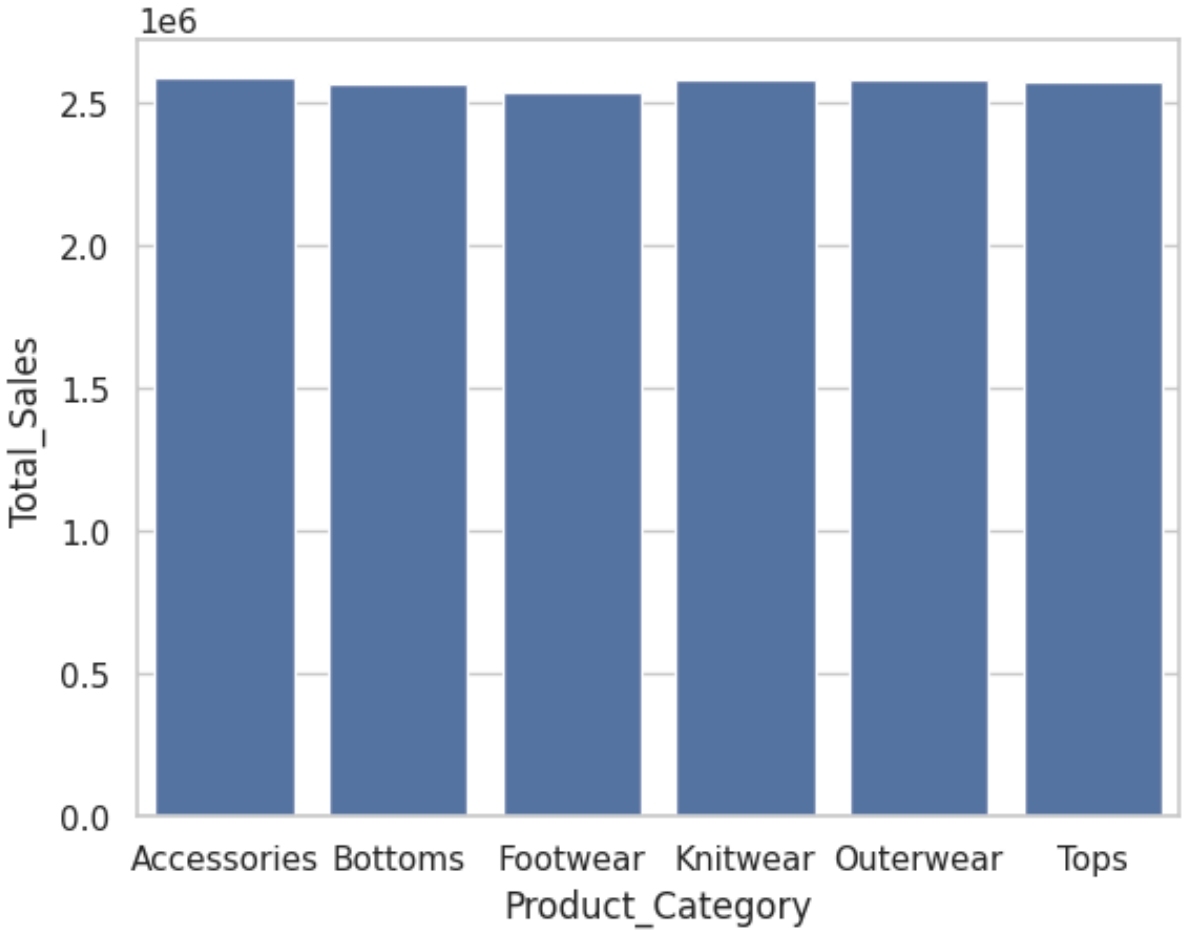


Figure 7.2: Revenue by product category

FINDING

The category range is narrow, so no category is supported as a single dominant revenue dependency.

BUSINESS MEANING

The category revenue outputs are close; Accessories is highest and Footwear is lowest.

DECISION / IMPLICATION

Use the result as a decision signal to test or investigate, not as proof of causality.

3. Revenue by City

BUSINESS QUESTION

Is revenue geographically concentrated?

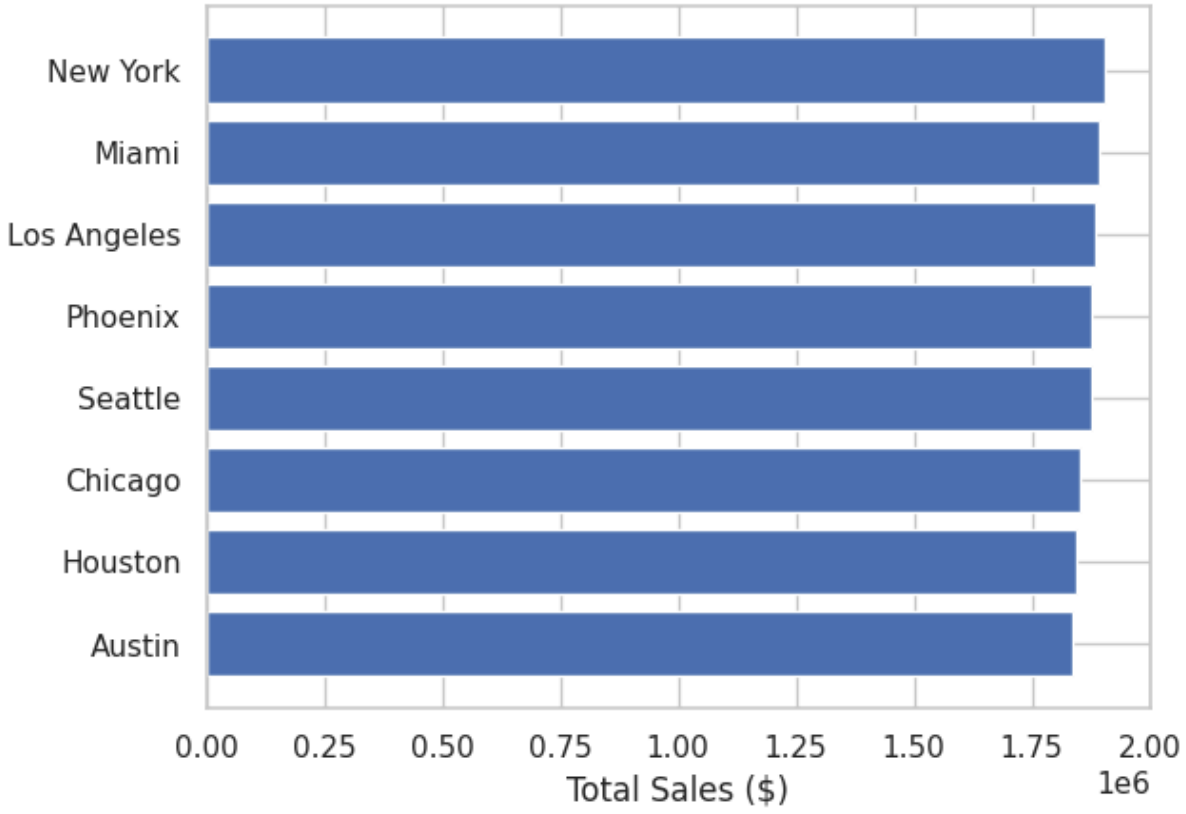


Figure 7.3: Revenue by city

FINDING

New York leads named cities and Austin is lowest; the gap is modest, so broad network improvement is better supported than a single-city rescue narrative.

BUSINESS MEANING

The notebook ranking excludes Unknown before ranking cities.

DECISION / IMPLICATION

Use the result as a decision signal to test or investigate, not as proof of causality.

4. Order Value Distribution

BUSINESS QUESTION

What does a typical order look like?

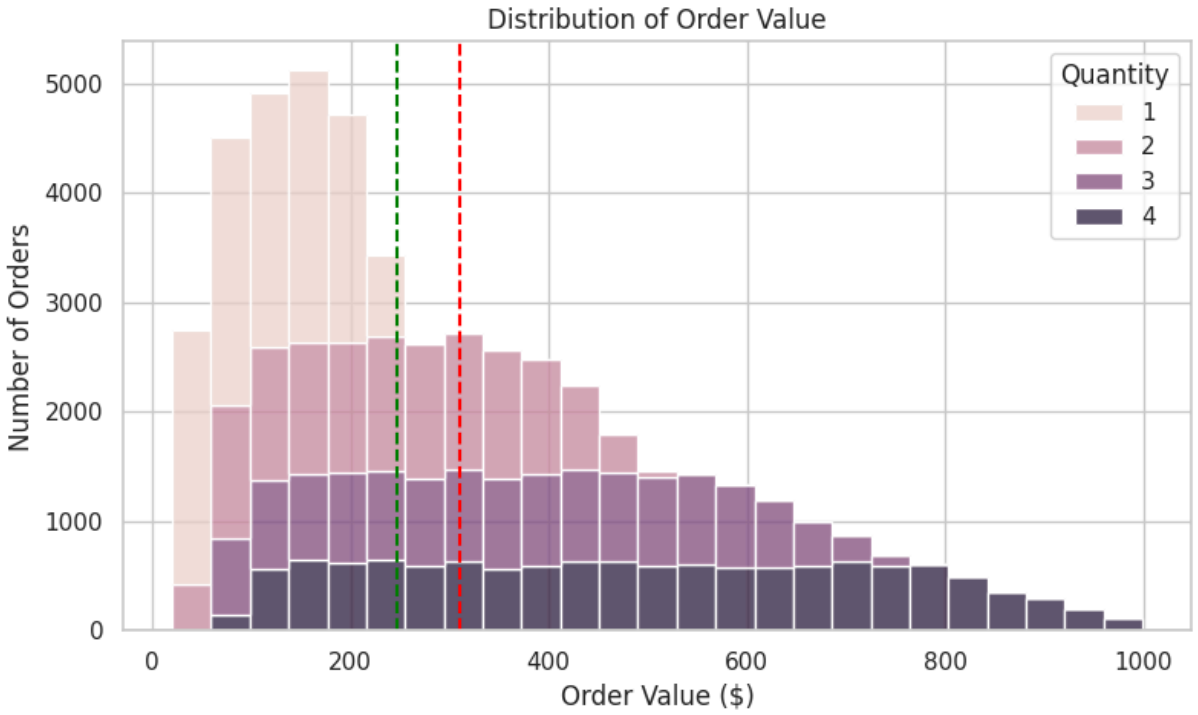


Figure 7.4: Order-value distribution by quantity

FINDING

The median is more representative of a typical order; a smaller high-value tail lifts the mean.

BUSINESS MEANING

The distribution is right-skewed: the median is 247.46 and the mean is 309.49.

DECISION / IMPLICATION

Keep high-value observations in the analysis and pair average metrics with median, IQR and quantity mix.

5. High-Value Order / Outlier Analysis

BUSINESS QUESTION

Are high-value orders unusual errors or legitimate baskets?

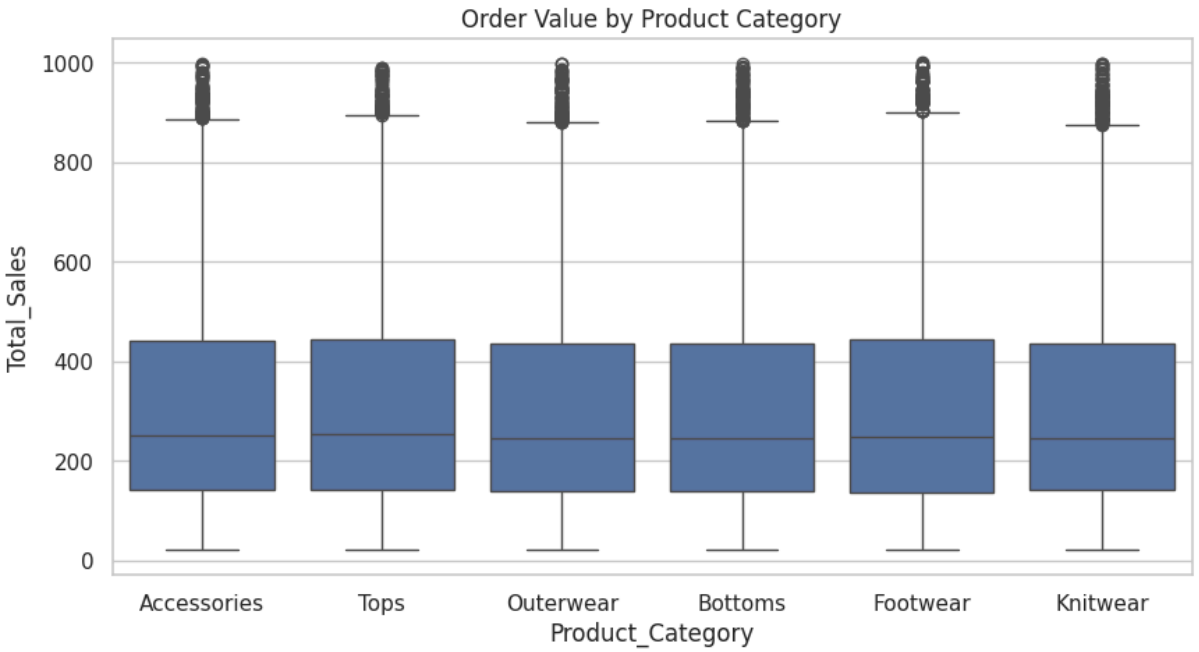


Figure 7.5: Order value by product category with outlier points

FINDING

The supplied analysis treats them as legitimate high-value baskets rather than automatic data errors; they are candidates for premium-basket testing.

BUSINESS MEANING

The IQR upper fence is 887.5825; 534 valid orders exceed it and all have Quantity = 4.

DECISION / IMPLICATION

Keep high-value observations in the analysis and pair average metrics with median, IQR and quantity mix.

6. Observed Price–Demand Relationship

BUSINESS QUESTION

Does unit price show an observed relationship with quantity?



Figure 7.6: Unit price versus order value, coloured by quantity

FINDING

No clear observed linear price–quantity relationship is detected. The chart shows order value increasing with price and visible quantity bands, but it does not prove a demand response.

BUSINESS MEANING

This is not a price-elasticity chart. The supplied evidence reports Pearson r ≈ 0.0037 and p ≈ 0.40 for Unit_Price versus Quantity.

DECISION / IMPLICATION

Use the result as a decision signal to test or investigate, not as proof of causality.

7. Correlation Heatmap

BUSINESS QUESTION

Which numerical variables move together?

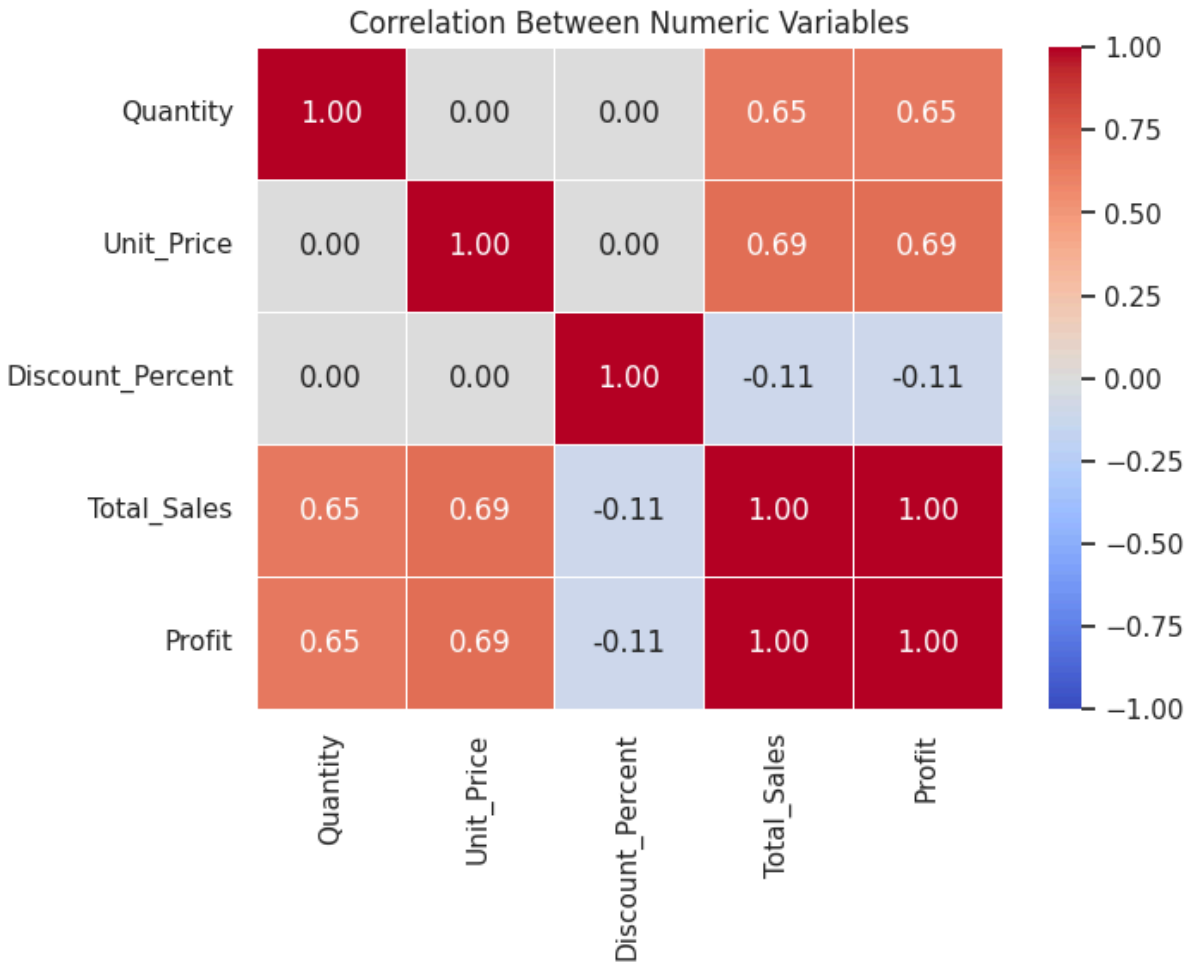


Figure 7.8: Correlation heatmap

FINDING

The heatmap supports careful association language. Profit should not be treated as an independent driver because it is derived from sales at approximately 45%.

BUSINESS MEANING

Quantity and Unit_Price are positively associated with Total_Sales; Discount_Percent is weakly negative; Profit is perfectly aligned with Total_Sales.

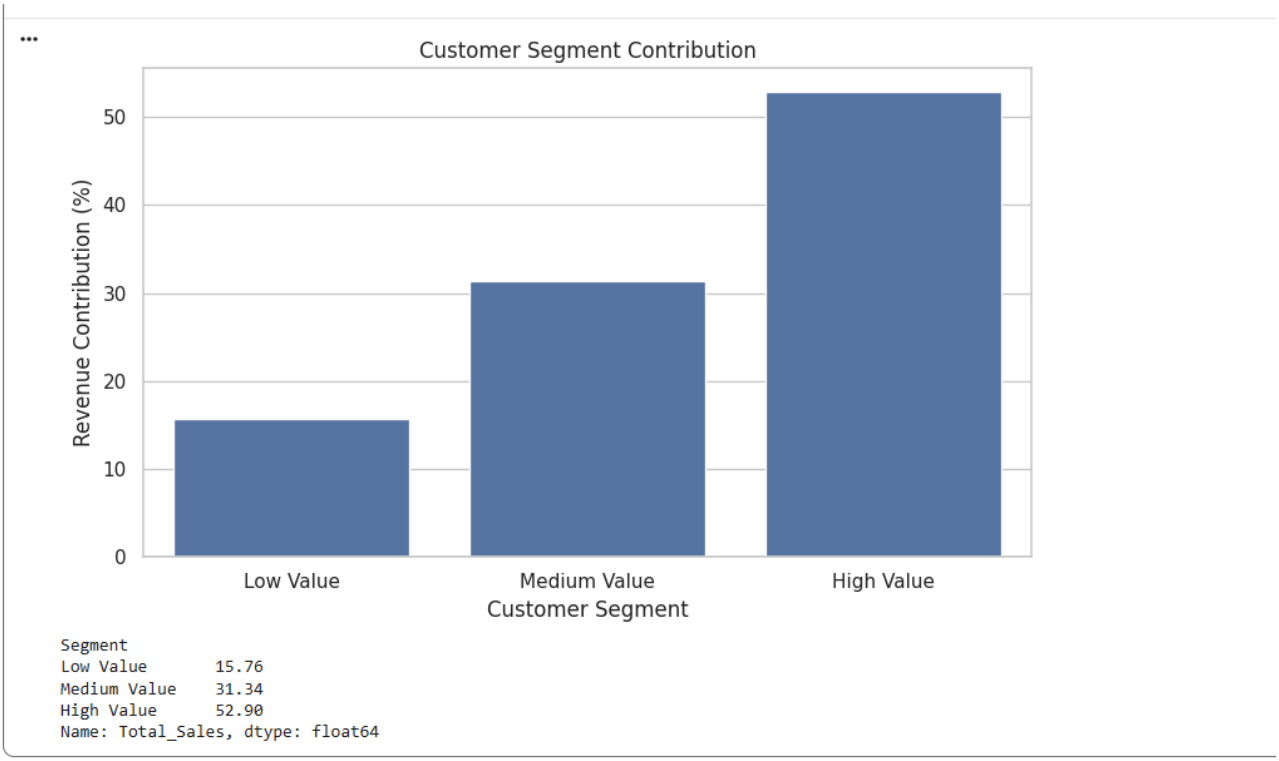
DECISION / IMPLICATION

Use the result as a decision signal to test or investigate, not as proof of causality.

9. Customer Segment Contribution

BUSINESS QUESTION

Where should customer-retention effort be focused?




```
Segment
Low Value       15.76
Medium Value    31.34
High Value      52.90
Name: Total_Sales, dtype: float64
```

Available evidence	Value
Customers	8,969
Top 10% revenue share	20.23%
Fixed $500 VIP rule	8,434 customers classified as High-Value VIP in the SQL output
Interpretation	Fixed-threshold VIP classification is too broad; percentile-based tiers are more useful

FINDING

The supplied evidence supports customer-retention prioritization using concentration and repeat-purchase behaviour, but no customer-segment chart image is available in the project files.

DECISION / IMPLICATION

Use percentile-based tiers and repeat-purchase behaviour rather than relying only on the fixed $500 threshold.

7.6 Python Findings

Finding	Evidence
Stable revenue	Monthly revenue range 1,233,697.49–1,322,879.71.
Balanced dimensions	Category, city and style revenue ranges are modest.
Discount association	AOV decreases from 338.77 at 0% discount to 275.93 at 20%; units/order remain near 2.5.
Legitimate high-value tail	534 valid orders exceed 887.5825; all have Quantity = 4.
Price–quantity relationship	r ≈ 0.0037; p ≈ 0.40; no clear relationship detected.

Finding	Evidence
Broad customer base	Top 10% contribute 20.23% of revenue; fixed $500 VIP rule is broad.

8. Data Modelling

8.1 Power BI Data Model

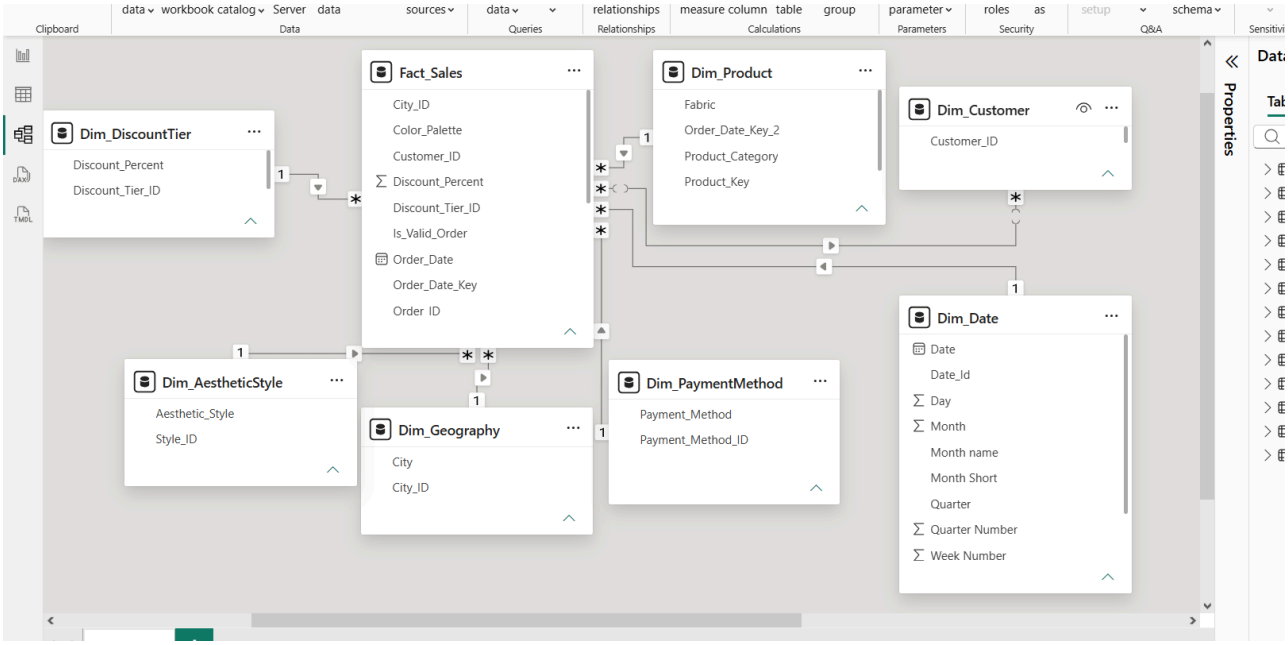
The PBIX model is organized around Fact_Sales with supporting date, product, aesthetic-style, customer, geography, payment-method and discount-tier dimensions. The model also contains _Measures and helper structures such as Waterfall_Stage and MetricMatrix.

8.2 Fact and Dimension Structure

Object	Role	Fields / usage
Fact_Sales	Fact table	Transaction-level sales, quantity, unit price, discount, profit, Customer_ID and Is_Valid_Order.
Dim_Date	Date dimension	Date, month, month name, month short, quarter and hierarchy fields.
Dim_Product	Product dimension	Product_Category, Fabric and Product_Label.
Dim_AestheticStyle	Style dimension	Aesthetic_Style.
Dim_Customer	Customer dimension	Customer_ID, Value Segment and Lifetime Sales bins.
Dim_Geography	Geography dimension	City.
Dim_PaymentMethod	Payment dimension	Payment_Method.
Dim_DiscountTier	Discount dimension	Discount_Percent and Tier Label.
_Measures	Measure table	Reusable KPI and analytical measures.
Waterfall_Stage / MetricMatrix	Helper structures	Waterfall and metric presentation support.

8.3 Relationships

Visual bindings show Fact_Sales being analysed through the listed dimensions. Exact relationship cardinalities and cross-filter directions were not available as readable text in the supplied PBIX extraction; this report documents the semantic structure without inventing relationship settings.



9. DAX Analysis

The PBIX visual bindings confirm the measure names below. The compressed DataModel stream did not expose exact DAX formula text as readable source, so formulas are not reconstructed here.

9.1 KPI Measures

Measure	Usage
Total Sales	Revenue cards, trends, category/city charts and decomposition tree.
Total Profit	Profit KPI and category analysis.
Profit Margin %	Executive profitability card/trend.
AOV	Average order value cards and basket analysis.
Total Orders	Order KPI.
Total Quantity	Units KPI.
Customer Count	Customer KPI.
Sales per Customer	Customer-value KPI.
Orders per Customer	Customer-frequency KPI.

9.2 Calculated Measures

Measure	Usage
Avg Daily Sales	Month-length normalized sales.
Avg Units/Order	Basket-size comparison.
Effective Discount Rate	Discount intensity.
Payment Channel Share %	Payment mix.
Sales MoM %	Month-over-month trend.
Q1 to Q4 Growth %	Quarter comparison.
High Value Sales Share %	Share of sales from high-value orders.
High Value Target %	Gauge target.

Measure	Usage
Deep Discount $	Deep-discount amount narrative.
Outliers / Invalid Rows / Total Rows	Data-quality KPI cards.

9.3 Analytical Measures

The model also exposes analytical fields including Is_Valid_Order, Price Band, Order_Value_Band, Value Segment, Lifetime Sales (bins), Tier Label, Waterfall Stage and date fields. These fields support filtering, segmentation, high-value order analysis, dashboard narratives and drilldown.

10. Power BI Dashboard

The PBIX contains exactly five report pages. The documentation preserves the existing names and summarizes the page purpose, visuals, filters and management use. Rendered dashboard screenshots were not available as image files, so manual insertion placeholders are retained.

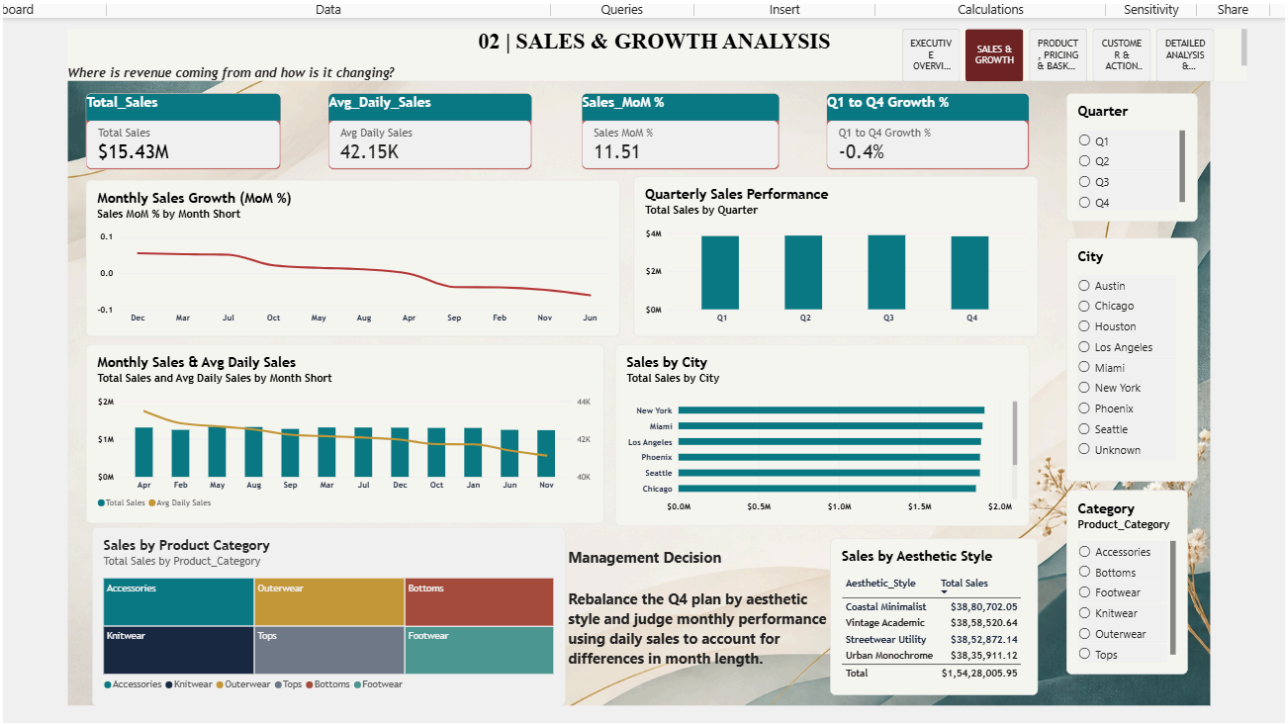
10.2 Executive Overview

Dashboard element	Documented content
Purpose	Is the business growing profitably?
Visuals	KPI cards for Total Sales, Total Profit, Profit Margin % and AOV; waterfall; sales/profit combo; margin/discount line; insight textbox; slicers; category table.
Filters	Quarter, Month and Product Category



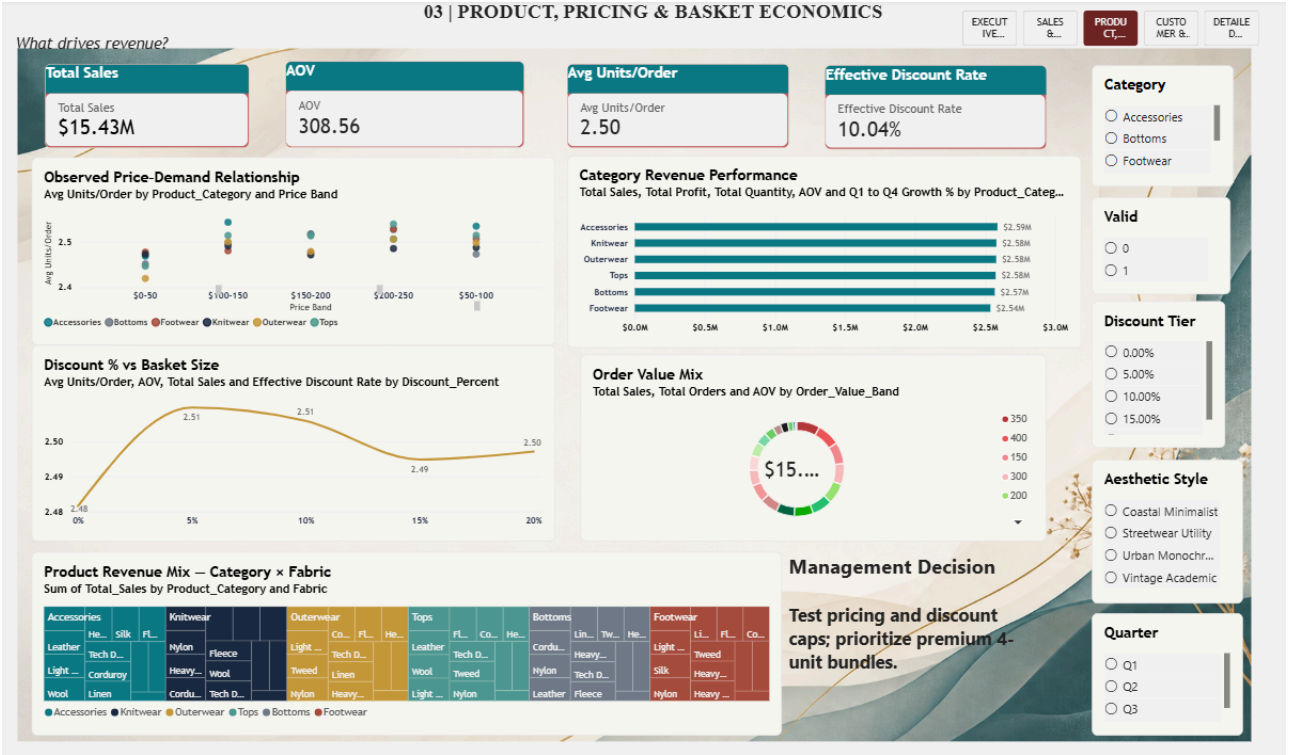
10.3 Sales & Growth

Dashboard element	Documented content
Purpose	Where is revenue coming from and how is it changing?
Visuals	Monthly sales and average daily sales combo; quarterly sales; Sales MoM %; city bar; category treemap; aesthetic-style table; KPI cards.
Filters	Quarter, Product Category and City



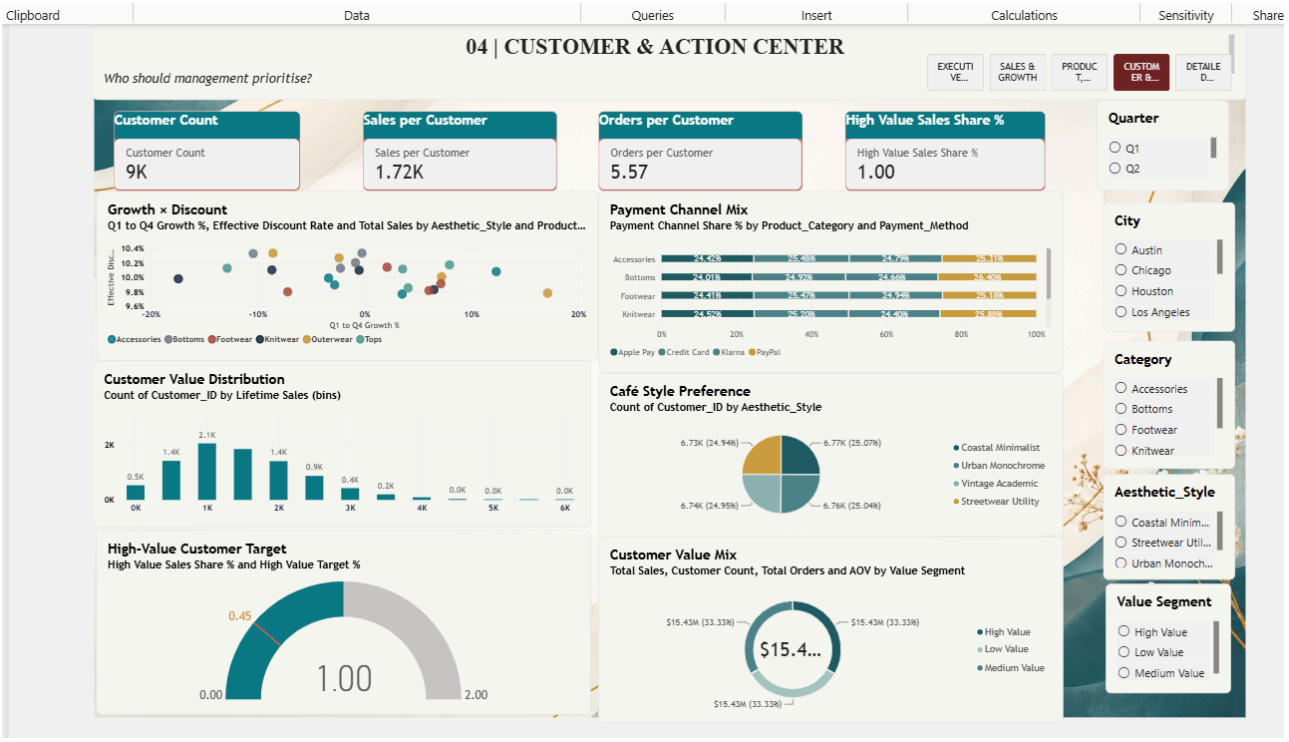
10.4 Product, Pricing & Basket Economics

Dashboard element	Documented content
Purpose	What drives revenue?
Visuals	Product/fabric treemap; category comparison; price-band scatter; discount trend; order-value donut; KPI cards.
Filters	Quarter, Aesthetic Style, Product Category, Discount Percent and Is_Valid_Order



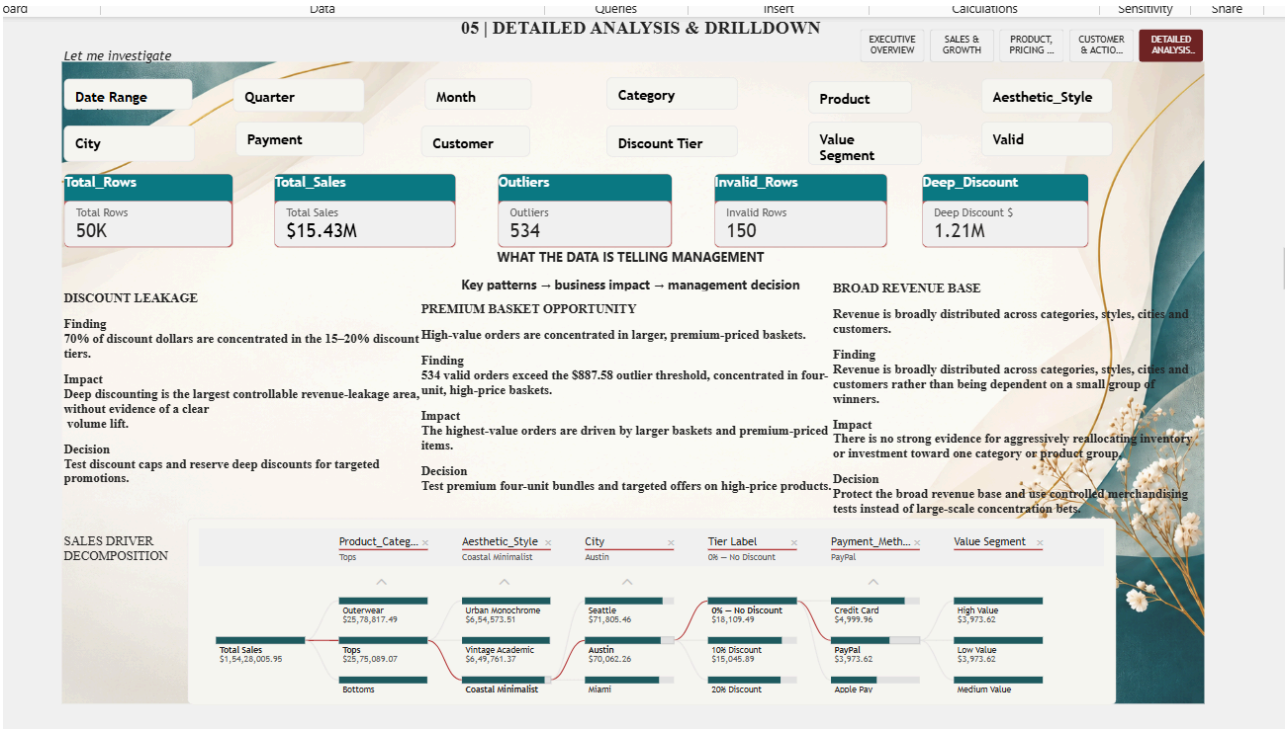
10.5 Customer & Action Center

Dashboard element	Documented content
Purpose	Who should management prioritise?
Visuals	Customer lifetime-sales bins; value-segment donut; segment/style chart; payment-share chart; growth/discount scatter; high-value gauge; customer KPIs.
Filters	Quarter, Aesthetic Style, Product Category, City and Value Segment



10.6 Detailed Analysis & Drilldown

Dashboard element	Documented content
Purpose	Let management investigate detailed drivers.
Visuals	Detailed slicers; Total Rows, Total Sales, Outliers, Invalid Rows and Deep Discount $ cards; insight textboxes; decomposition tree.
Filters	Quarter, style, category, city, month, payment, customer, product, discount, value segment and validity



11. Analysis and Findings

Theme	Observed finding	Evidence
Revenue trend	2024 revenue is relatively stable.	Monthly valid revenue 1.23M–1.32M.
Category concentration	No category dominates.	Accessories 2.588M; Footwear 2.537M.
Geographic concentration	Named cities are broadly balanced.	New York 1.905M; Austin 1.834M.
Discounts and baskets	Deeper discounts associate with lower AOV, not a visible units lift.	AOV 338.77 at 0% vs 275.93 at 20%; units/order ≈2.5.
Price and quantity	No clear observed relationship.	r ≈0.0037; p ≈0.40.
High-value orders	High-value tail is legitimate in supplied analysis.	534 orders above 887.5825; all Quantity = 4.
Customer concentration	Broad customer base.	Top 10% contribute 20.23% of revenue.
Profit	Not independent for margin ranking.	Profit = 45% of Total_Sales in valid data.

12. Business Insights

Insight	Business interpretation
Basket economics	Because categories/cities are balanced, pricing, discount depth and basket composition are more actionable levers than rescuing one category.
Discount leakage	The supplied PBIX narrative states 70% of discount dollars are concentrated in the 15–20% tiers; deeper discounts should be tested rather than assumed to lift volume.
Premium baskets	Four-unit, high-value orders are an observable opportunity for bundle testing.
Customer targeting	A fixed $500 VIP rule classifies most customers as VIP; percentile-based tiers and repeat purchase are more discriminating.
Data quality	Unknown City, Unknown Fabric and imputed café values weaken targeting and should be improved at source.

13. Business Recommendations

Area	Recommendation	Evidence / guardrail
Pricing	Run controlled price tests; do not infer price effects from observational data.	Unit price/quantity r ≈0.0037; p≈0.40.
Discount	Consider a standard cap around 10%; reserve 15–20% for targeted promotions; test AOV and units/order.	Deeper discount levels associate with lower AOV without visible unit lift; no causal claim.
Product / basket	Test bundles or multi-item offers around four-unit premium baskets.	534 valid IQR flags are all Quantity = 4.
Customer	Replace fixed $500 VIP rule with percentile-based tiers; use repeat purchase behaviour.	8,434 of 8,969 customers are classified VIP by the fixed threshold.
Data quality	Improve city/café-style capture and prevent zero/negative quantities at source.	Unknown fields and zero/negative quantity issues are documented.
Profitability	Use actual cost data if true margin analysis is required.	Profit is mechanically derived at approximately 45% of sales.

14. Technical Notes and Known Issues

Issue	Documentation treatment
CSV vs notebook filename	Notebook references an Excel file; attached cleaned artifact is CSV.
SQL vs Python population	SQL includes 50,000 cleaned rows; Python excludes 150 zero-quantity rows.
Fabric label mismatch	SQL script specifies Unspecified; cleaned CSV shows Unknown for the imputed Fabric group.
Date type	SQL schema stores Order_Date as VARCHAR; explicit conversion is safer for production.
Profit derivation	Profit is exactly 45% of sales; avoid independent margin claims.
DAX text	Measure names/bindings are visible; exact formulas are not readable in extracted PBIX metadata.
Dashboard screenshots	Page definitions are documented; rendered screenshots require manual insertion.

15. Limitations

- The dataset is described in the notebook as synthetic-looking and unusually balanced.
- No real company, operating context or source-system definition is supplied.
- Observational associations do not establish causation.
- No experiment, forecasting or machine-learning model is included.
- Missing/imputed values limit city, fabric and lifestyle interpretation.
- Fixed profit construction limits independent margin analysis.
- Customer-segment visual output is not present as a rendered image in the supplied files.

16. Design Standards and Colour Palette

Element	Standard
Page / layout	A4 portrait; clean white background; restrained spacing.
Typography	Aptos/Segoe UI style; Arial used for Word compatibility.
Headings	Navy main headings; orange section headings; blue subheadings.

Element	Standard
Tables	Light-green headers; thin blue-grey borders; compact readable text.
Charts	Only supplied notebook output images; no generated replacement visuals.
Code	Consolas in light-grey blocks; selected snippets in main report.
Colours	#0D3A74 primary; #1A73E8 analytical blue; #DD7030 documentation accent; #DDECCF table headers; #FFFFFF background.

17. End-to-End Technical Workflow

Step	Workflow
1	Inspect raw CSV shape, schema, nulls, duplicates, date text and numeric ranges.
2	Create SQL database and raw table.
3	Trim, standardize, impute, recalculate and deduplicate.
4	Persist cleaned output and validate row counts.
5	Parse dates and create Python valid-order population.
6	Compute descriptive statistics, correlations and outlier flags.
7	Present exactly nine business visualizations with evidence and decisions.
8	Model Fact_Sales and dimensions in Power BI.
9	Apply the actual Power Query data preparation and transformation steps in Power BI before loading the data into the data model.
10	Deliver five-page dashboard, findings, insights and recommendations.

18. Conclusion

The project provides a connected analytics story from raw data to management decision support. SQL resolves data-quality issues and creates reusable analysis tables. Python validates the cleaned population and shows that 2024 revenue is stable, categories and cities are broadly balanced, discounts are associated with lower order value without a clear unit lift, price and quantity show no clear observed relationship, and high-value four-unit orders form a legitimate tail.

Power BI then organizes the data into a fact-and-dimension model with KPI and analytical measures across five report pages. The practical value is a disciplined basis for testing discount caps, evaluating premium baskets, improving customer segmentation, and fixing source-data quality. The report intentionally avoids causal claims and documents the limitations of the supplied synthetic-looking data and mechanically derived profit field.

19. Appendix

19.A Complete SQL Artifacts

The complete SQL scripts supplied with the project are reproduced below as technical references. The main report uses selected excerpts and outputs; this appendix keeps the full implementation available without interrupting the business narrative.

19.A.1 Data_Cleaning_using_SQL.sql

```
CREATE DATABASE IF NOT EXISTS capstone_aesthetic_db;
USE capstone_aesthetic_db;

DROP TABLE IF EXISTS raw_aesthetic_sales;

CREATE TABLE raw_aesthetic_sales (
    Order_ID VARCHAR(50),
    Order_Date VARCHAR(50),
    Customer_ID VARCHAR(50),
    Product_Category VARCHAR(50),
    Fabric VARCHAR(50),
    Aesthetic_Style VARCHAR(50),
    Color_Palette VARCHAR(100),
    Preferred_Cafe_Style VARCHAR(100),
    Quantity INT,
    Unit_Price DECIMAL(10,2),
    Discount_Percent DECIMAL(5,2),
    Total_Sales DECIMAL(10,2),
    Profit DECIMAL(10,2),
    City VARCHAR(50),
    Payment_Method VARCHAR(50)
);


-- Step 0: Disable Safe Updates to allow mass UPDATE operations
SET SQL_SAFE_UPDATES = 0;

-- STEP 1: TRIM WHITESPACE (TEXT / VARCHAR COLUMNS ONLY)
-- Removes accidental leading and trailing spaces from text fields

UPDATE raw_aesthetic_sales
SET
    Order_ID             = TRIM(Order_ID),
    Order_Date           = TRIM(Order_Date),
    Customer_ID          = TRIM(Customer_ID),
    Product_Category     = TRIM(Product_Category),
    Fabric               = TRIM(Fabric),
    Aesthetic_Style      = TRIM(Aesthetic_Style),
    Color_Palette        = TRIM(Color_Palette),
    Preferred_Cafe_Style = TRIM(Preferred_Cafe_Style),
    City                 = TRIM(City),
    Payment_Method       = TRIM(Payment_Method);


-- STEP 2: STANDARDIZE TEXT CASING


UPDATE raw_aesthetic_sales
SET Fabric = CONCAT(UPPER(SUBSTRING(Fabric, 1, 1)), LOWER(SUBSTRING(Fabric, 2)))
WHERE Fabric IS NOT NULL AND Fabric != '';

UPDATE raw_aesthetic_sales
SET Product_Category = CONCAT(UPPER(SUBSTRING(Product_Category, 1, 1)), LOWER(SUBSTRING(Product_Category, 2)))
WHERE Product_Category IS NOT NULL AND Product_Category != '';

UPDATE raw_aesthetic_sales
SET City = CONCAT(UPPER(SUBSTRING(City, 1, 1)), LOWER(SUBSTRING(City, 2)))
WHERE City IS NOT NULL AND City != '';


-- STEP 3: IMPUTE MISSING AND NULL TEXT VALUES

UPDATE raw_aesthetic_sales
SET Fabric = 'Unspecified'
WHERE Fabric IS NULL OR Fabric = '';

UPDATE raw_aesthetic_sales
SET City = 'Unknown'
WHERE City IS NULL OR City = '';

UPDATE raw_aesthetic_sales
SET Preferred_Cafe_Style = 'Unspecified'
WHERE Preferred_Cafe_Style IS NULL OR Preferred_Cafe_Style = '';
```

```
-- Replaces negative quantities (-1) with 0 and recalculates total sales
UPDATE raw_aesthetic_sales
SET Quantity = CASE
    WHEN Quantity < 0 THEN 0
    ELSE Quantity
END;

UPDATE raw_aesthetic_sales
SET Total_Sales = ROUND(Quantity * Unit_Price * (1 - Discount_Percent), 2),
    Profit = ROUND(Total_Sales * 0.45, 2);


-- STEP 5: REMOVE DUPLICATE RECORDS USING A CTE & WINDOW FUNCTION


DROP TABLE IF EXISTS cleaned_aesthetic_sales;

CREATE TABLE cleaned_aesthetic_sales AS
WITH RankedOrders AS (
    SELECT *,
           ROW_NUMBER() OVER (
               PARTITION BY Order_ID
               ORDER BY Order_Date
           ) AS row_num
    FROM raw_aesthetic_sales
)
SELECT
    Order_ID,
    Order_Date,
    Customer_ID,
    Product_Category,
    Fabric,
    Aesthetic_Style,
    Color_Palette,
    Preferred_Cafe_Style,
    Quantity,
    Unit_Price,
    Discount_Percent,
    Total_Sales,
    Profit,
    City,
    Payment_Method
FROM RankedOrders
WHERE row_num = 1;


-- STEP 6: VERIFY CLEANED DATASET RECORD COUNT

SELECT COUNT(*) AS total_cleaned_records FROM cleaned_aesthetic_sales;

-- Re-enable Safe Updates
SET SQL_SAFE_UPDATES = 1;
```

19.A.2 Data_Analysis_SQL.sql

```
USE capstone_aesthetic_db;
-- Objective: Inspect the first few rows of the cleaned table to verify schema and column values.
SELECT
    Order_ID,
    Order_Date,
    Customer_ID,
    Product_Category,
    Total_Sales,
    Profit
FROM cleaned_aesthetic_sales
LIMIT 10;

-- Objective: Extract high-value orders where Total_Sales exceeds $200 and the payment method is Credit Card.
SELECT
    Order_ID,
    Customer_ID,
    Product_Category,
    Total_Sales,
    Payment_Method
FROM cleaned_aesthetic_sales
WHERE Total_Sales > 200.00
  AND Payment_Method = 'Credit Card'
ORDER BY Total_Sales DESC;

-- Objective: List all unique product categories and aesthetic style combinations available in the catalog.

SELECT DISTINCT
    Product_Category,
    Aesthetic_Style,
    Fabric
FROM cleaned_aesthetic_sales
ORDER BY Product_Category, Aesthetic_Style;

-- Objective: Calculate Total Orders, Total Revenue, Total Profit, and Overall Average Order Value (AOV).
SELECT
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    COUNT(DISTINCT Customer_ID) AS Total_Customers,
    ROUND(SUM(Total_Sales), 2) AS Gross_Revenue,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND(AVG(Total_Sales), 2) AS Overall_AOV
FROM cleaned_aesthetic_sales;

-- Objective: Group performance metrics by product category to find top-selling apparel lines.
SELECT
    Product_Category,
    COUNT(Order_ID) AS Total_Orders,
    SUM(Quantity) AS Total_Units_Sold,
    ROUND(SUM(Total_Sales), 2) AS Category_Revenue,
    ROUND(SUM(Profit), 2) AS Category_Profit
FROM cleaned_aesthetic_sales
GROUP BY Product_Category
ORDER BY Category_Revenue DESC;

-- Objective: Find cities generating over $5,000 in revenue and filter out low-volume markets.
SELECT
    City,
    COUNT(Order_ID) AS Total_Orders,
    ROUND(SUM(Total_Sales), 2) AS City_Revenue
FROM cleaned_aesthetic_sales
GROUP BY City
HAVING SUM(Total_Sales) > 5000.00
ORDER BY City_Revenue DESC;

-- Objective: Break down total revenue, profit, and profit margin percentage across aesthetic styles (e.g., Cottagecore,
Y2K, Dark Academia).

SELECT
    Aesthetic_Style,
    COUNT(Order_ID) AS Order_Count,
    ROUND(SUM(Total_Sales), 2) AS Total_Revenue,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Total_Sales)) * 100, 2) AS Profit_Margin_Pct
FROM cleaned_aesthetic_sales
GROUP BY Aesthetic_Style
ORDER BY Total_Revenue DESC;

-- Objective: Analyze interaction patterns between lifestyle traits (Preferred_Cafe_Style) and product choices (Fabric).
SELECT
    Preferred_Cafe_Style,
    Fabric,
    COUNT(Order_ID) AS Total_Purchases,
    ROUND(SUM(Total_Sales), 2) AS Total_Revenue
FROM cleaned_aesthetic_sales
GROUP BY Preferred_Cafe_Style, Fabric
ORDER BY Preferred_Cafe_Style, Total_Purchases DESC;

--  Objective: Categorize customers into VIP, Mid-Tier, and Low-Spend tiers based on total spend.
```

```sql
SELECT
    Customer_ID,
    COUNT(Order_ID) AS Lifetime_Orders,
    ROUND(SUM(Total_Sales), 2) AS Lifetime_Spend,
    CASE
        WHEN SUM(Total_Sales) >= 500 THEN 'High-Value VIP'
        WHEN SUM(Total_Sales) BETWEEN 200 AND 499.99 THEN 'Mid-Tier Customer'
        ELSE 'Casual Shopper'
    END AS Customer_Tier
FROM cleaned_aesthetic_sales
GROUP BY Customer_ID
ORDER BY Lifetime_Spend DESC;
-- Objective: Evaluate how discount tiers (No Discount, 1–10%, >10%) influence order quantities.
SELECT
    CASE
        WHEN Discount_Percent = 0 THEN '0% (Full Price)'
        WHEN Discount_Percent BETWEEN 0.01 AND 0.10 THEN '1% - 10% Discount'
        ELSE 'Over 10% Discount'
    END AS Discount_Band,
    COUNT(Order_ID) AS Total_Orders,
    ROUND(AVG(Quantity), 2) AS Avg_Units_Per_Order,
    ROUND(SUM(Total_Sales), 2) AS Total_Revenue
FROM cleaned_aesthetic_sales
GROUP BY
    CASE
        WHEN Discount_Percent = 0 THEN '0% (Full Price)'
        WHEN Discount_Percent BETWEEN 0.01 AND 0.10 THEN '1% - 10% Discount'
        ELSE 'Over 10% Discount'
    END
ORDER BY Total_Revenue DESC;

-- Objective: Pivot payment preferences across aesthetic styles using conditional aggregation.
SELECT
    Aesthetic_Style,
    COUNT(CASE WHEN Payment_Method = 'Credit Card' THEN 1 END) AS Credit_Card_Orders,
    COUNT(CASE WHEN Payment_Method = 'PayPal' THEN 1 END) AS PayPal_Orders,
    COUNT(CASE WHEN Payment_Method = 'UPI' THEN 1 END) AS UPI_Orders,
    COUNT(CASE WHEN Payment_Method = 'Cash on Delivery' THEN 1 END) AS COD_Orders
FROM cleaned_aesthetic_sales
GROUP BY Aesthetic_Style
ORDER BY Aesthetic_Style;


-- Objective: Extract monthly time trends using DATE_FORMAT()
SELECT
    DATE_FORMAT(Order_Date, '%Y-%m') AS Order_Month,
    COUNT(Order_ID) AS Monthly_Orders,
    ROUND(SUM(Total_Sales), 2) AS Monthly_Revenue,
    ROUND(SUM(Profit), 2) AS Monthly_Profit
FROM cleaned_aesthetic_sales
GROUP BY DATE_FORMAT(Order_Date, '%Y-%m')
ORDER BY Order_Month ASC;

-- Objective: Identify individual orders where revenue higher than the overall average revenue per order.
SELECT
    Order_ID,
    Customer_ID,
    Product_Category,
    Total_Sales
FROM cleaned_aesthetic_sales
WHERE Total_Sales > (
    SELECT AVG(Total_Sales)
    FROM cleaned_aesthetic_sales
)
ORDER BY Total_Sales DESC;

-- Objective: Rank top product categories within every aesthetic style independently.
WITH RankedCategories AS (
    SELECT
        Aesthetic_Style,
        Product_Category,
        ROUND(SUM(Total_Sales), 2) AS Category_Revenue,
        DENSE_RANK() OVER (
            PARTITION BY Aesthetic_Style
            ORDER BY SUM(Total_Sales) DESC
        ) AS Category_Rank
    FROM cleaned_aesthetic_sales
    GROUP BY Aesthetic_Style, Product_Category
)
SELECT
    Aesthetic_Style,
    Product_Category,
    Category_Revenue,
    Category_Rank
FROM RankedCategories
WHERE Category_Rank <= 3;

-- Objective: Calculate running total revenue chronologically across dates.
SELECT
    Order_Date,
```

```
    ROUND(SUM(Total_Sales), 2) AS Daily_Revenue,
    ROUND(SUM(SUM(Total_Sales)) OVER (
        ORDER BY Order_Date
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
    ), 2) AS Cumulative_Revenue
FROM cleaned_aesthetic_sales
GROUP BY Order_Date
ORDER BY Order_Date ASC;

-- Objective: Use LAG() to calculate days elapsed between consecutive purchases for repeat customers.
WITH CustomerOrders AS (
    SELECT
        Customer_ID,
        Order_ID,
        Order_Date,
        LAG(Order_Date, 1) OVER (
            PARTITION BY Customer_ID
            ORDER BY Order_Date
        ) AS Previous_Order_Date
    FROM cleaned_aesthetic_sales
)
SELECT
    Customer_ID,
    Order_ID,
    Order_Date,
    Previous_Order_Date,
    DATEDIFF(Order_Date, Previous_Order_Date) AS Days_Between_Orders
FROM CustomerOrders
WHERE Previous_Order_Date IS NOT NULL
ORDER BY Customer_ID, Order_Date;

SELECT * FROM cleaned_aesthetic_sales  WHERE Order_Date IS NULL;

-- Objective: Aggregate city sales first, then join city metrics back to high-value order counts
WITH CitySummary AS (
    SELECT
        City,
        COUNT(Order_ID) AS Total_City_Orders,
        ROUND(SUM(Total_Sales), 2) AS City_Revenue,
        ROUND(AVG(Total_Sales), 2) AS City_AOV
    FROM cleaned_aesthetic_sales
    GROUP BY City
),
HighValueCityOrders AS (
    SELECT
        City,
        COUNT(Order_ID) AS High_Value_Orders
    FROM cleaned_aesthetic_sales
    WHERE Total_Sales >= 250.00
    GROUP BY City
)
SELECT
    cs.City,
    cs.Total_City_Orders,
    COALESCE(hv.High_Value_Orders, 0) AS High_Value_Orders,
    cs.City_Revenue,
    cs.City_AOV
FROM CitySummary cs
LEFT JOIN HighValueCityOrders hv ON cs.City = hv.City
ORDER BY cs.City_Revenue DESC;

-- Objective: Calculate cumulative percentage of total sales contributed by customers ranked by spend.
WITH CustomerTotals AS (
    SELECT
        Customer_ID,
        ROUND(SUM(Total_Sales), 2) AS Customer_Spend
    FROM cleaned_aesthetic_sales
    GROUP BY Customer_ID
),
RankedCustomers AS (
    SELECT
        Customer_ID,
        Customer_Spend,
        SUM(Customer_Spend) OVER (ORDER BY Customer_Spend DESC) AS Cumulative_Spend,
        SUM(Customer_Spend) OVER () AS Total_Business_Spend
    FROM CustomerTotals
)
SELECT
    Customer_ID,
    Customer_Spend,
    Cumulative_Spend,
    ROUND((Cumulative_Spend / Total_Business_Spend) * 100, 2) AS Cumulative_Revenue_Pct
FROM RankedCustomers
ORDER BY Customer_Spend DESC;

-- Objective: Compute each category's revenue percentage contribution to its respective city's total revenue.
WITH CityCategorySales AS (
    SELECT
        City,
        Product_Category,
```

```
        ROUND(SUM(Total_Sales), 2) AS Category_Revenue
    FROM cleaned_aesthetic_sales
    GROUP BY City, Product_Category
)
SELECT
    City,
    Product_Category,
    Category_Revenue,
    SUM(Category_Revenue) OVER (PARTITION BY City) AS Total_City_Revenue,
    ROUND((Category_Revenue / SUM(Category_Revenue) OVER (PARTITION BY City)) * 100, 2) AS Pct_Contribution_To_City
FROM CityCategorySales
ORDER BY City, Pct_Contribution_To_City DESC;

-- Objective: Calculate month-over-month revenue growth percentage.

WITH MonthlySales AS (
    SELECT
        DATE_FORMAT(Order_Date, '%Y-%m') AS Sales_Month,
        ROUND(SUM(Total_Sales), 2) AS Current_Month_Revenue
    FROM cleaned_aesthetic_sales
    GROUP BY DATE_FORMAT(Order_Date, '%Y-%m')
),
MonthlyGrowth AS (
    SELECT
        Sales_Month,
        Current_Month_Revenue,
        LAG(Current_Month_Revenue, 1) OVER (ORDER BY Sales_Month) AS Previous_Month_Revenue
    FROM MonthlySales
)
SELECT
    Sales_Month,
    Current_Month_Revenue,
    Previous_Month_Revenue,
    ROUND(((Current_Month_Revenue - Previous_Month_Revenue) / Previous_Month_Revenue) * 100, 2) AS MoM_Growth_Pct
FROM MonthlyGrowth;
```

19.B Python Notebook Reference

Complete Python code remains in python_data_visualization_9_charts.ipynb. The main report shows only the loading/filtering snippet and the final nine chart outputs. The notebook includes the documented inspection, descriptive statistics, grouping, correlations, IQR outlier analysis and final story.

```
# Core valid-order population used by the supplied notebook
df = df[df['Quantity'] > 0].copy()
df['Order_Month'] = df['Order_Date'].dt.to_period('M').astype(str)
df['Month_Num'] = df['Order_Date'].dt.month

# IQR outlier rule used in the notebook
q1 = df['Total_Sales'].quantile(0.25)
q3 = df['Total_Sales'].quantile(0.75)
upper_fence = q3 + 1.5 * (q3 - q1)
```

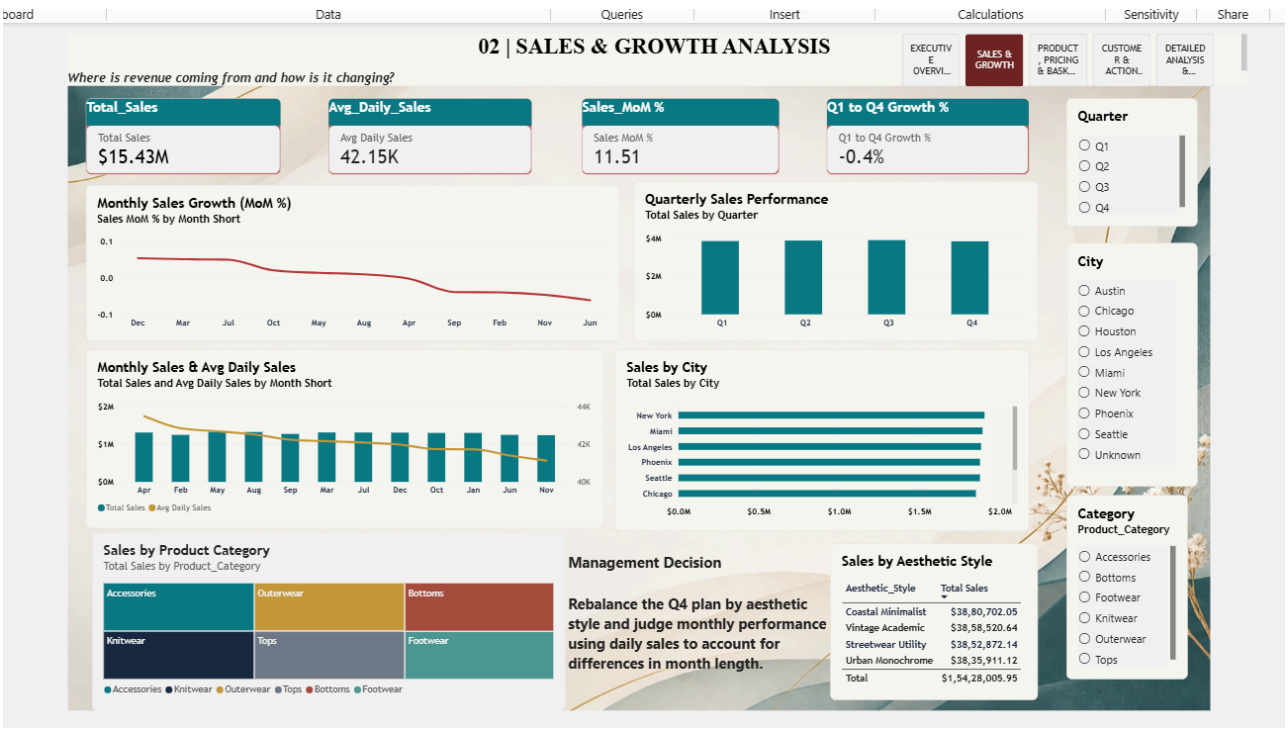
19.C Supporting Technical Tables

Artifact	What it supports
new_raw_dataset.csv	Raw quality counts and source schema.
new_cleaned_data_power.csv	Cleaned record counts and reconciled metrics.
major_project.pbix	Five page names, visual bindings, model objects and report theme.
python_data_visualization_9_charts.ipynb	Nine supplied chart images and statistical narrative.

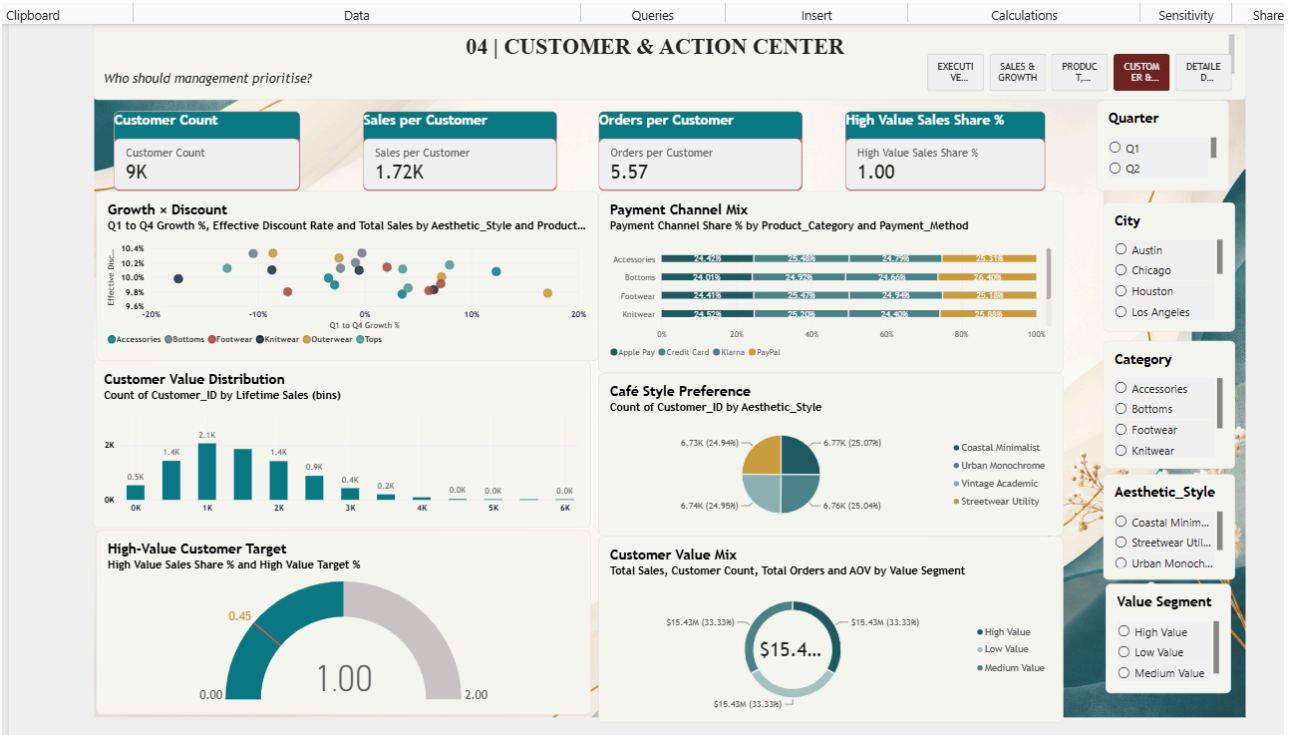
19.D Screenshot Insertion Checklist

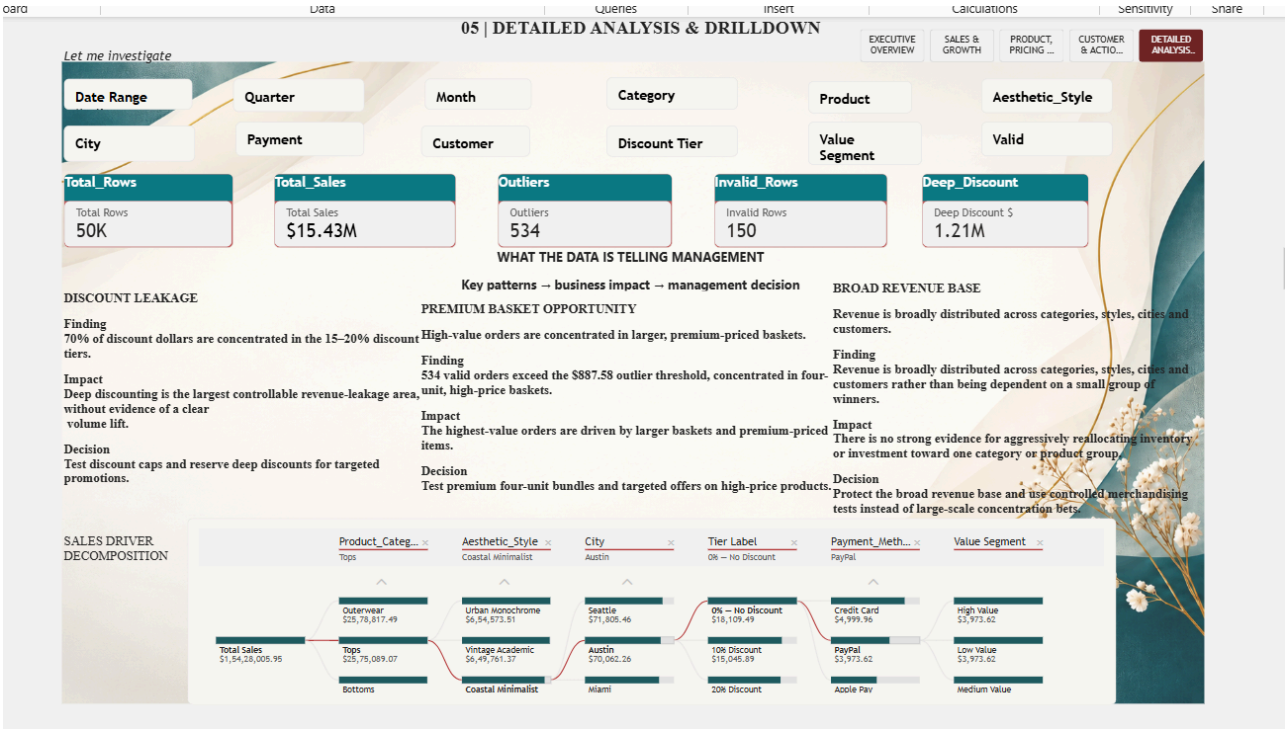
Insert only images captured from the supplied project files. Do not generate replacement dashboard screenshots.


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Share
1 | Executive Overview
EXECUTIVE OVERVIEW
SALES & GROWTH
PRODUCT, PRICING &...
CUSTOMER & ACTION...
DETAILED ANALYSIS &...
Is the business growing profitably?
Total Sales
Total Sales
$15.43M
Total Profit
Total Profit
$6.94M
Profit Margin
Profit Margin %
45.0%
Average Order Value
AOV
308.56
Monthly Sales & Profit Trend.
Total Sales and Total Profit by Month Short
$1.0M
$0.5M
$0.0M
Sep Oct Nov May Mar Jun Jul Jan Feb Dec Aug Apr
Total Sales Total Profit
Revenue Waterfall
Waterfall Value by Waterfall Stage
30M
20M
10M
0M
Gross Revenue
Net Sales
Discount Amount
Total
Increase Decrease Total
Month name
April
August
December
February
January
July
Quarter
Q1
Q2
Q3
Q4
Profit Margin vs Effective Discount Rate
Profit Margin % and Effective Discount Rate by Month Short
45%
45%
10.4%
10.2%
10.0%
9.8%
Apr Aug Dec Feb Jan Jul Jun Mar May Nov Oct Sep
Profit Margin % Effective Discount Rate
MANAGEMENT INSIGHT
Revenue is stable across 2024, while profit margin remains broadly flat at ~45%. Discounting is the main commercial lever, with an effective discount rate of ~10%. Management should focus on controlling discount depth while protecting sales volume and high-value customer demand.
Category
Product_Category
Accessories
Bottoms
Footwear
Knitwear
Outerwear
Tops


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02 | SALES & GROWTH ANALYSIS
EXECUTIVE OVERVI...
SALES & GROWTH
PRODUCT, PRICING & BASK...
CUSTOMER & ACTION...
DETAILED ANALYSIS &...
Where is revenue coming from and how is it changing?
Total_Sales
Total Sales
$15.43M
Avg_Daily_Sales
Avg Daily Sales
42.15K
Sales_MoM %
Sales MoM %
11.51
Q1 to Q4 Growth %
Q1 to Q4 Growth %
-0.4%
Monthly Sales Growth (MoM %)
Sales MoM % by Month Short
0.1
0.0
-0.1
Dec Mar Jul Oct May Aug Apr Sep Feb Nov Jun
Quarterly Sales Performance
Total Sales by Quarter
$4M
$2M
$0M
Q1 Q2 Q3 Q4
Monthly Sales & Avg Daily Sales
Total Sales and Avg Daily Sales by Month Short
$2M
$1M
$0M
44K
42K
40K
Apr Feb May Aug Sep Mar Jul Dec Oct Jan Jun Nov
Total Sales Avg Daily Sales
Sales by City
Total Sales by City
New York
Miami
Los Angeles
Phoenix
Seattle
Chicago
$0.0M $0.5M $1.0M $1.5M $2.0M
Quarter
Q1
Q2
Q3
Q4
City
Austin
Chicago
Houston
Los Angeles
Miami
New York
Phoenix
Seattle
Unknown
Category
Product_Category
Accessories
Bottoms
Footwear
Knitwear
Outerwear
Tops
Sales by Product Category
Total Sales by Product_Category
Accessories
Outerwear
Bottoms
Knitwear
Tops
Footwear
Accessories Knitwear Outerwear Tops Bottoms Footwear
Management Decision
Rebalance the Q4 plan by aesthetic style and judge monthly performance using daily sales to account for differences in month length.
Sales by Aesthetic Style
Aesthetic_Style
Total Sales
Coastal Minimalist $38,80,702.05
Vintage Academic $38,58,520.64
Streetwear Utility $38,52,872.14
Urban Monochrome $38,35,911.12
Total $1,54,28,005.95


03 | PRODUCT, PRICING & BASKET ECONOMICS
What drives revenue?
EXECUTIVE...
SALES &...
PRODUCT,...
CUSTOMER &...
DETAILED...
Total Sales
Total Sales
$15.43M
AOV
AOV
308.56
Avg Units/Order
Avg Units/Order
2.50
Effective Discount Rate
Effective Discount Rate
10.04%
Observed Price-Demand Relationship
Avg Units/Order by Product_Category and Price Band
Avg Units/Order
Price Band
$0-50
$100-150
$150-200
$200-250
$50-100
Accessories Bottoms Footwear Knitwear Outerwear Tops
Category Revenue Performance
Total Sales, Total Profit, Total Quantity, AOV and Q1 to Q4 Growth % by Product_Categ...
Accessories $2.59M
Knitwear $2.58M
Outerwear $2.58M
Tops $2.58M
Bottoms $2.57M
Footwear $2.54M
$0.0M $0.5M $1.0M $1.5M $2.0M $2.5M $3.0M
Discount % vs Basket Size
Avg Units/Order, AOV, Total Sales and Effective Discount Rate by Discount_Percent
Order Value Mix
Total Sales, Total Orders and AOV by Order_Value_Band
$15....
350
400
150
300
200
Product Revenue Mix — Category × Fabric
Sum of Total_Sales by Product_Category and Fabric
Accessories Knitwear Outerwear Tops Bottoms Footwear
Management Decision
Test pricing and discount caps; prioritize premium 4-unit bundles.
Category
Accessories
Bottoms
Footwear
Valid
0
1
Discount Tier
0.00%
5.00%
10.00%
15.00%
Aesthetic Style
Coastal Minimalist
Streetwear Utility
Urban Monochr...
Vintage Academic
Quarter
Q1
Q2
Q3


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04 | CUSTOMER & ACTION CENTER
Who should management prioritise?
EXECUTIVE...
SALES & GROWTH
PRODUCT,...
CUSTOMER &...
DETAILED...
Customer Count
Customer Count
9K
Sales per Customer
Sales per Customer
1.72K
Orders per Customer
Orders per Customer
5.57
High Value Sales Share %
High Value Sales Share %
1.00
Growth × Discount
Q1 to Q4 Growth %, Effective Discount Rate and Total Sales by Aesthetic_Style and Product...
Effective Dis...
Q1 to Q4 Growth %
Accessories Bottoms Footwear Knitwear Outerwear Tops
Payment Channel Mix
Payment Channel Share % by Product_Category and Payment_Method
Accessories
Bottoms
Footwear
Knitwear
Apple Pay Credit Card Klarna PayPal
Customer Value Distribution
Count of Customer_ID by Lifetime Sales (bins)
Café Style Preference
Count of Customer_ID by Aesthetic_Style
6.73K (24.94%)
6.77K (25.07%)
6.74K (24.95%)
6.76K (25.04%)
Coastal Minimalist
Urban Monochrome
Vintage Academic
Streetwear Utility
High-Value Customer Target
High Value Sales Share % and High Value Target %
0.45
1.00
0.00
2.00
Customer Value Mix
Total Sales, Customer Count, Total Orders and AOV by Value Segment
$15.43M (33.33%)
$15.43M (33.33%)
$15.4...
$15.43M (33.33%)
High Value
Low Value
Medium Value
Quarter
Q1
Q2
City
Austin
Chicago
Houston
Los Angeles
Category
Accessories
Bottoms
Footwear
Knitwear
Aesthetic_Style
Coastal Minim...
Streetwear Util...
Urban Monoch...
Value Segment
High Value
Low Value
Medium Value


oard
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05 | DETAILED ANALYSIS & DRILLDOWN
EXECUTIVE OVERVIEW
SALES & GROWTH
PRODUCT, PRICING ...
CUSTOMER & ACTIO...
DETAILED ANALYSIS..
Let me investigate
Date Range
Quarter
Month
Category
Product
Aesthetic_Style
City
Payment
Customer
Discount Tier
Value Segment
Valid
Total_Rows
Total Rows
50K
Total_Sales
Total Sales
$15.43M
Outliers
Outliers
534
Invalid_Rows
Invalid Rows
150
Deep_Discount
Deep Discount $
1.21M
WHAT THE DATA IS TELLING MANAGEMENT
Key patterns → business impact → management decision
DISCOUNT LEAKAGE
Finding
70% of discount dollars are concentrated in the 15–20% discount tiers.
Impact
Deep discounting is the largest controllable revenue-leakage area, without evidence of a clear volume lift.
Decision
Test discount caps and reserve deep discounts for targeted promotions.
PREMIUM BASKET OPPORTUNITY
High-value orders are concentrated in larger, premium-priced baskets.
Finding
534 valid orders exceed the $887.58 outlier threshold, concentrated in four-unit, high-price baskets.
Impact
The highest-value orders are driven by larger baskets and premium-priced items.
Decision
Test premium four-unit bundles and targeted offers on high-price products.
BROAD REVENUE BASE
Revenue is broadly distributed across categories, styles, cities and customers.
Finding
Revenue is broadly distributed across categories, styles, cities and customers rather than being dependent on a small group of winners.
Impact
There is no strong evidence for aggressively reallocating inventory or investment toward one category or product group.
Decision
Protect the broad revenue base and use controlled merchandising tests instead of large-scale concentration bets.
SALES DRIVER DECOMPOSITION
Product_Categ...
Tops
Aesthetic_Style
Coastal Minimalist
City
Austin
Tier Label
0% – No Discount
Payment_Meth...
PayPal
Value Segment
Total Sales
$1,54,28,005.95
Tops
$25,75,089.07
Outerwear
$25,78,817.49
Bottoms
Urban Monochrome
$6,54,573.51
Vintage Academic
$6,49,761.37
Coastal Minimalist
Seattle
$71,805.46
Austin
$70,062.26
Miami
0% – No Discount
$18,109.49
10% Discount
$15,045.89
20% Discount
Credit Card
$4,999.96
PayPal
$3,973.62
Apple Pay
High Value
$3,973.62
Low Value
$3,973.62
Medium Value